

Exhibit 95

U.S. Patent No. 10,217,523 Domestic Industry Claim Chart

EXHIBIT 95

Domestic Industry Claim Chart for Claims of U.S. Patent No. 10,217,523 (the “’523 Patent”) Based on Practicing ’523 Products

As shown in the chart below, the Practicing ’523 Products, which include Netlist DDR5 LP MRDIMMs, SK [REDACTED], and SK hynix MRDIMMs, practice at least claim 1 of the ’523 patent. It should be understood that Netlist’s contentions are not limited to these products but instead encompass any and all reasonably similar products and components that practice the claims in the same or similar manner as discussed below. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions by Respondents that the Practicing ’523 Products do not practice the ’523 patent as to any one or more of the limitations in the claim(s) below.

To the extent Respondents contend that any differences are alleged to exist between the asserted claims below and the Practicing ’523 Products, such differences are insubstantial and the Practicing ’523 Products perform substantially the same function, in substantially the same way, to yield substantially the same result, and therefore the Practicing ’523 Products practice the claims below under the doctrine of equivalents. Netlist may supplement, modify or amend its contentions under the doctrine of equivalents based on any claim constructions, additional information obtained through formal discovery or other means, and/or based on other circumstances including any assertions that the Practicing ’523 Products do not satisfy the technical domestic industry by Respondents as to any one or more of the limitations in the claims below.

[REDACTED]

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Claim 1	Evidence of Use
[1.a] A memory module accessible in a computer system by a system memory controller via a system memory bus, comprising:	SK hynix's DDR5 MRDIMM meets all of the limitations of claim 1. To the extent that the preamble is limiting, SK hynix's DDR5 MRDIMM is "[a] memory module accessible in a computer system by a system memory controller via a system memory bus." Below are images of the front and back of a SK hynix DDR5 MRDIMM, which is a Multiplexed Rank Dual In-line Memory Module.  The images show the front and back of a SK hynix DDR5 MRDIMM. The front view (top) features a white label with the following text: SK hynix KOREA, CE, DDR5 EDS MRDIMM, 32GB 2Rx8 PC5-88/548-MD0-1312-XT, PN HMC688BDJHA464N AA, SN 80AD01251014FF1B95, and a barcode. The back view (bottom) shows the green PCB with multiple black memory chips and gold-plated pins.

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Ex. CC-36 (Product Images) at 13, 14, 16.

First, SK hynix's MRDIMMs are each a dual in-line memory module.

For example, MRDIMM stands for "Multiplexed Rank Dual In-line Memory Module." *See, e.g.*, Ex. CC-42 (SK hynix MRDIMM) at 1; *see also, e.g.*, Ex. CC-62 (Montage HK Proof) at 43 ("MRDIMM" is defined as "multiplexed rank dual in-line memory module . . .").

As another example, SK hynix's DDR5 MRDIMM utilizes components made by Montage Technology, including Montage's DDR5 Multiplexed Rank Registering Clock Driver (MRCD) and Multiplexed Rank Data Buffer (MDB) chips.

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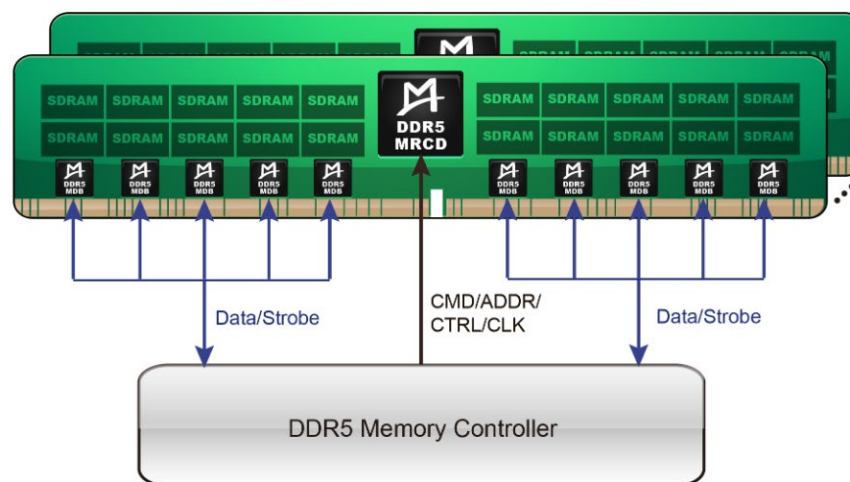
Ex. CC-36 (Product Images) at 14, 15.

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Montage states that its “MRCD chip is applied in next-generation DDR5 MRDIMM / MCRDIMM memory modules[.]” See Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1 (emphasis added); Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer).

General Description

The M88MR5RCD01 is a first-generation DDR5 Multiplexed Rank Registering Clock Driver (MRCD) chip that supports data rates of up to 8800MT/s. This chip buffers and re-drives the address, command, clock, and control signals from the memory controller. Compared to traditional DDR5 RCD chips, the MRCD can generate four independent chip-select signals at standard rates, enabling more flexible and efficient memory management operations. Working in conjunction with Multiplexed Rank Data Buffer (MDB) chips, the MRCD chip is applied in next-generation DDR5 MRDIMM / MCRDIMM memory modules, effectively doubling memory system bandwidth to meet the urgent demand for memory throughput in applications such as deep learning and large-scale data analysis.



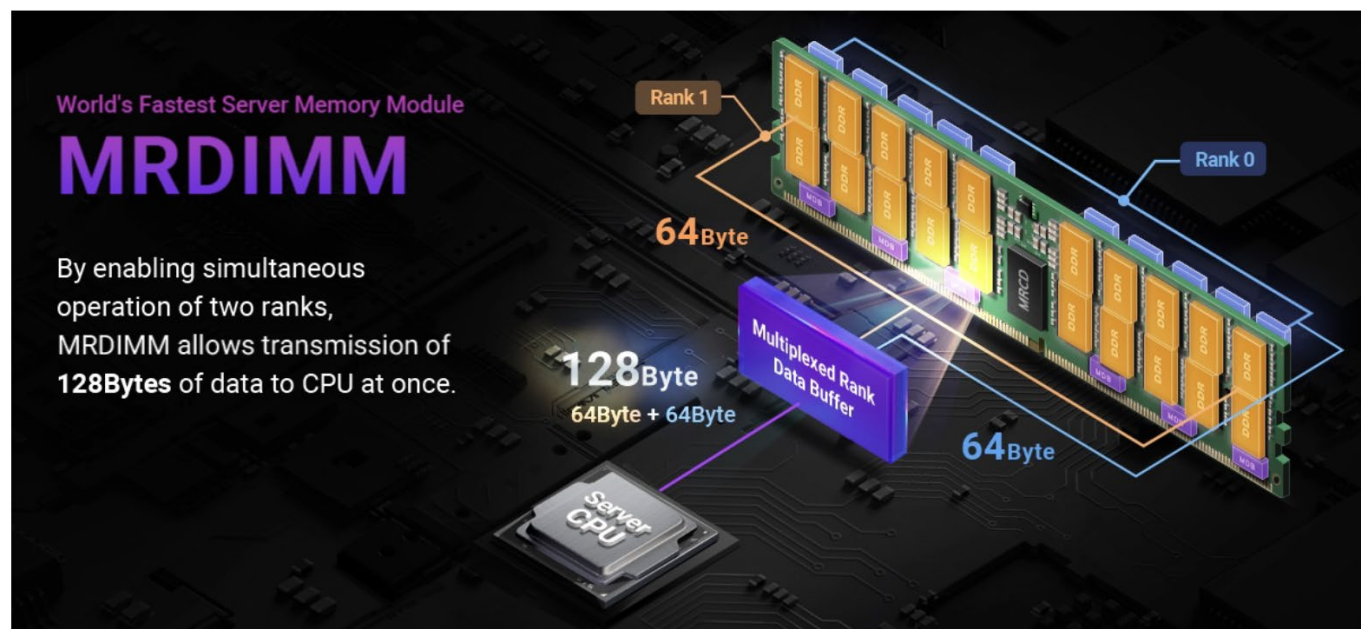
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	Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1–2. Second, the MRDIMM memory module is accessible in a computer system by a system memory controller via a system memory bus (e.g., as a pathway between a CPU and DRAMs for data communication, which would go through a memory slot in the case of communication between a memory controller and a DIMM). For example, SK hynix advertises on its website, with animation, the following (Ex. CC-42 (SK hynix MRDIMM) at 1–2):
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How MRDIMM Works

By facilitating simultaneous operation of two ranks, MRDIMM enables the transmission of 128 bytes of data to the CPU at once, compared to the 64 bytes typically fetched by conventional DRAM modules. This increase in the amount of data sent to the CPU each time supports twice the speed of a single DRAM.



According to Ex. CC-63 (SK hynix Blog Post) at 3:

Multiplexed Rank Dual In-line Memory Module (MRDIMM): A server memory module product in which multiple DRAM chips are mounted on a substrate. Speed is enhanced by operating two ranks—the basic

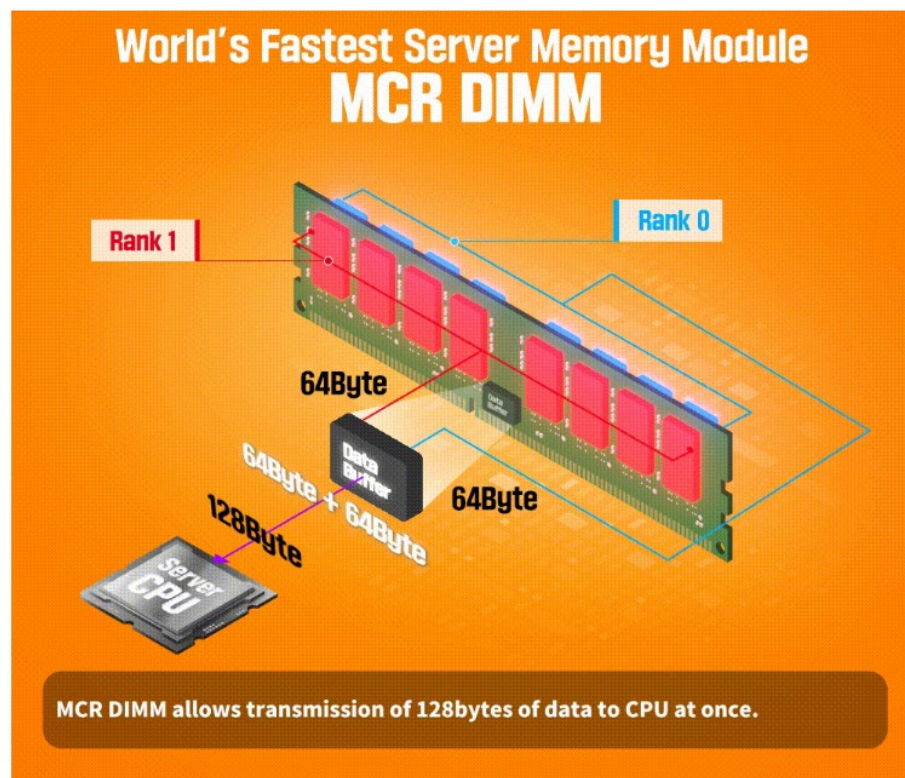
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operating units of the module—simultaneously.

See also Ex. CC-64 (SK hynix Newsroom Interview) at 2–3:

SK hynix Pushes Technological Limits to Lead the Server DRAM Market

The MCR DIMM represents a leap forward in performance, typically unseen in the DRAM sector. It is common to see an operating speed increase of about 800 Mbps (Megabits per second) when developing a new generation of the DDR5 DRAM. For example, if the DRAM offered an operating speed of 4,800 Mbps, the next generation is expected to have a speed of approximately 5,600 Mbps. However, SK hynix made yet another breakthrough by developing the MCR DIMM that boasts an operating speed of over 8,000 Mbps.



▲ Figure 3. The operation structure of the world's fastest server DRAM, the MCR DIMM.

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	<i>See also</i> Ex. CC-50 (TrendForce News MRDIMM-MCRDIMM) (referencing MRDIMM and MCRDIMM interchangeably). Ex. CC-64 (SK hynix Newsroom Interview) at 1 (“MCR DIMM (Multiplexer Combined Ranks Dual In-line Memory Module): A module product with multiple DRAMs bonded to a motherboard, in which two ranks² (basic information processing units) operate simultaneously, resulting in improved speed.”; “²Rank: A set of basic data transfer units exported from the DRAM module to the CPU. Usually, 64 bytes of data are transferred to the CPU as a unit.”):
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SK hynix once again proved to be the first and the best when it comes to innovations as it became the world's first company to [successfully develop working samples of the DDR5 MCR DIMM](#)^{*}, a server DRAM product, in December 2022. This successful development of a sample significantly enhances the performance of the server DRAM. The biggest improvement is that the MCR DIMM breaks away from the traditional idea of focusing solely on the operating speed of individual DRAM units to boost the speed of DDR5. Instead, it adds a customized module to individual DRAM units to increase the speed. The operating speed of the MCR DIMM is over 8 Gbps (gigabits per second), which is at least 80% faster than existing server DRAMs that have a speed of 4.8 Gbps. This makes it the fastest server DRAM in the world.

^{*}MCR DIMM (Multiplexer Combined Ranks Dual In-line Memory Module): A module product with multiple DRAMs bonded to a motherboard, in which two ranks^{*} (basic information processing units) operate simultaneously, resulting in improved speed.

^{*}Rank: A set of basic data transfer units exported from the DRAM module to the CPU. Usually, 64 bytes of data are transferred to the CPU as a unit.

As SK hynix sees the server memory market as pivotal to overcoming the recent downturn of the semiconductor industry, the company has focused on developing technologies related to server DRAMs. The SK hynix team who achieved innovation through new methods and paths emphasized the importance of challenging oneself, saying: "No matter how difficult it is, opening up new paths is always worthwhile." This was the motivation for the MCR DIMM team members, Kim Hong Bae, Kim Yeong Jun, and Yi Jong Yun, who committed themselves to developing the leading server DRAM.

See also Ex. CC-65 (SK hynix Develops MCR DIMM) at 1 ("SK hynix designed the product in a way that enables simultaneous operation of two ranks by utilizing the data buffer³ installed onto the MCR DIMM"; "³Buffer: A component that optimizes signal transmission performance between DRAM and CPU. Mainly installed onto modules for servers requiring high performance and reliability.");

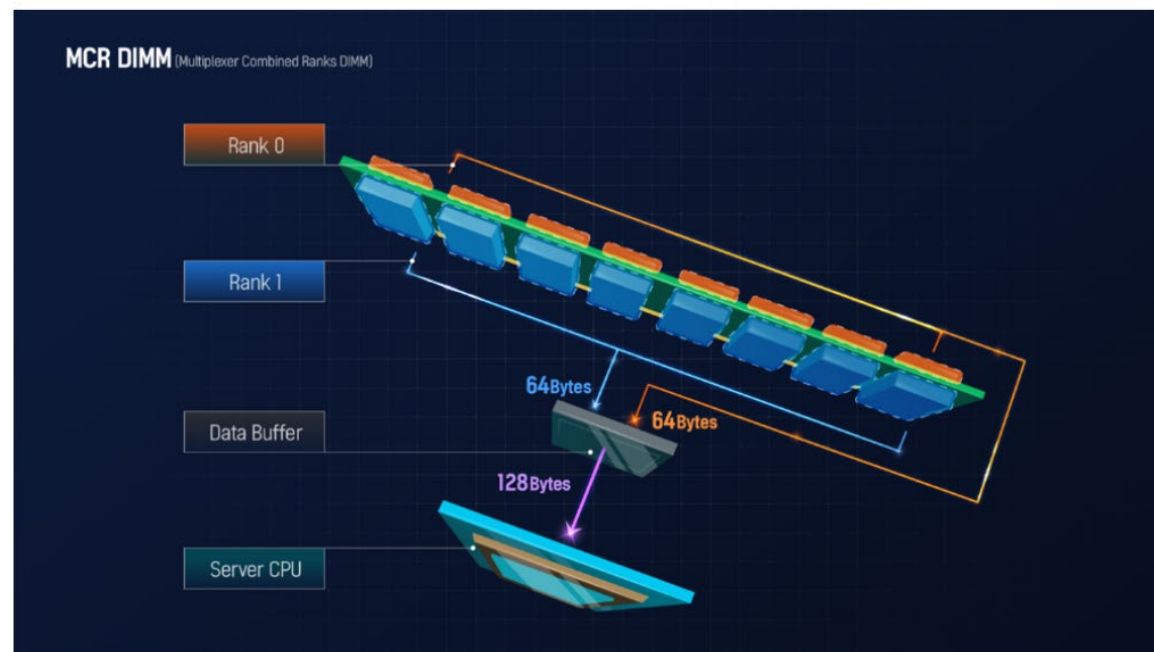
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	Seoul, December 8, 2022 SK hynix Inc. (or “the company”, www.skhynix.com) announced today that it has developed working samples of DDR5* Multiplexer Combined Ranks* (MCR) Dual In-line Memory Module, the world’s fastest server DRAM product. The new product has been confirmed to operate at the data rate of minimum 8Gbps, and at least 80% faster than 4.8Gbps of the existing DDR5 products. MCR DIMM is an achievement coming from out-of-the-box thinking with an aim to improve the operation speed of DDR5. Challenging the prevailing concept that the operation speed of DDR5 relies on that of DRAM chip itself, engineers sought to find a way to improve the speed of modules instead of chips for development of the latest product. SK hynix designed the product in a way that enables simultaneous operation of two ranks by utilizing the data buffer* installed onto the MCR DIMM based on Intel’s MCR technology. <i>*Double Data Rate (DDR), a DRAM standard mainly used for servers and client applications, has been developed up to the fifth generation. The MCR DIMM is a module product with multiple DRAM chips attached to the board and improved speed as a result of two ranks operating simultaneously.</i> <i>*Rank: A collection of basic transfer units of the data sent to CPU from DRAM module. A rank typically refers to 64 bytes of data to be transferred to central processing unit as a bundle.</i>
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*Buffer: A component that optimizes signal transmission performance between DRAM and CPU. Mainly installed onto modules for servers requiring high performance and reliability.

By enabling simultaneous operation of two ranks, MCR DIMM allows transmission of 128 bytes of data to CPU at once, compared with 64 bytes fetched generally in conventional DRAM module. An increase in the amount of data sent to the CPU each time supports the data transfer rate of minimum 8Gbps, twice as fast as a single DRAM.



This shows that MRDIMMs/MCRDIMMs are designed to be accessible in a computer system by a system memory controller in a CPU.

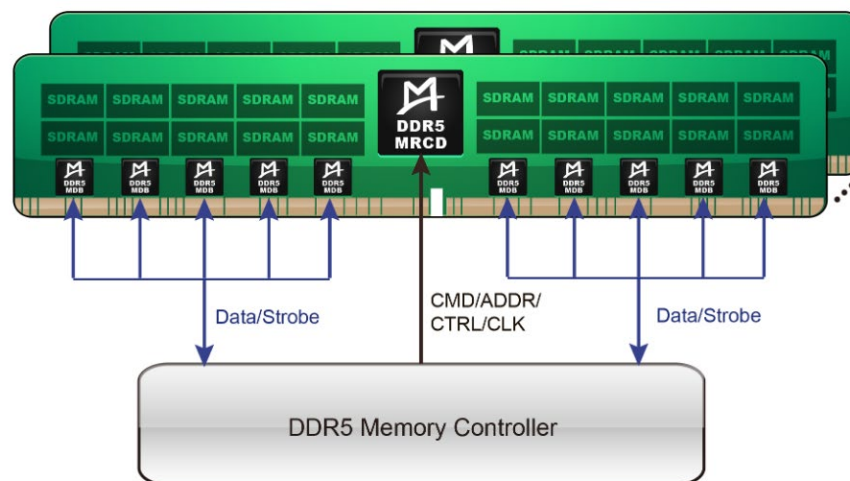
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As another example, Montage describes its MRCD chip as performing the function of “buffer[ing] and re-driv[ing] the address, command, clock, and control signals *from the memory controller*” to “support[] data rates of up to 8800MT/s.” Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1 (emphasis added); Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer).

General Description

The M88MR5RCD01 is a first-generation DDR5 Multiplexed Rank Registering Clock Driver (MRCD) chip that supports data rates of up to 8800MT/s. This chip buffers and re-drives the address, command, clock, and control signals from the memory controller. Compared to traditional DDR5 RCD chips, the MRCD can generate four independent chip-select signals at standard rates, enabling more flexible and efficient memory management operations. Working in conjunction with Multiplexed Rank Data Buffer (MDB) chips, the MRCD chip is applied in next-generation DDR5 MRDIMM / MCRDIMM memory modules, effectively doubling memory system bandwidth to meet the urgent demand for memory throughput in applications such as deep learning and large-scale data analysis.

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Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1–2 (showing data being exchanged between MRDIMM and DDR5 memory controller).

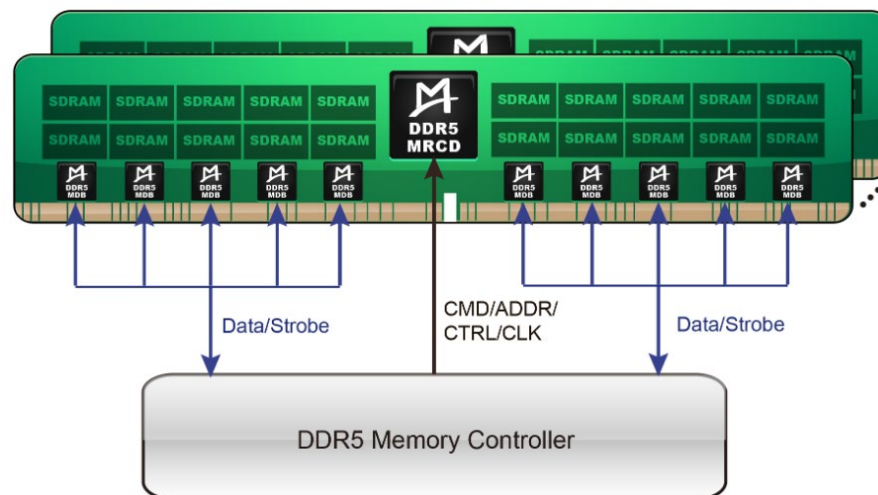
Likewise, the web page for Montage’s MDB states the “MDB chip buffers and re-drives data signals from the memory controller or DRAM devices” and it “efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal.” Ex. CC-22 (Montage M88MR5DB01) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer):

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General Description

The M88MR5DB01 is a Gen1 DDR5 Multiplexed Data Buffer (MDB) supporting data rates up to 8800MT/s. The MDB chip buffers and re-drives data signals from the memory controller or DRAM devices. It incorporates a dual-nibble host interface that operates at twice the speed of the DRAM interface. The DRAM side features four nibbles, with two allocated to each pseudo-channel. The MDB efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal.

Designed to work in conjunction with the Multiplexed Registering Clock Driver (MRCD), the MDB plays a crucial role in DDR5 MRDIMMs / MCRDIMMs. Each MRDIMM / MCRDIMM is equipped with one MRCD and ten MDBs. This configuration enables the MRDIMM / MCRDIMM module to concurrently access two DRAM memory ranks through the MDB chips, resulting in a doubling of bandwidth to address the demands of high-load applications.



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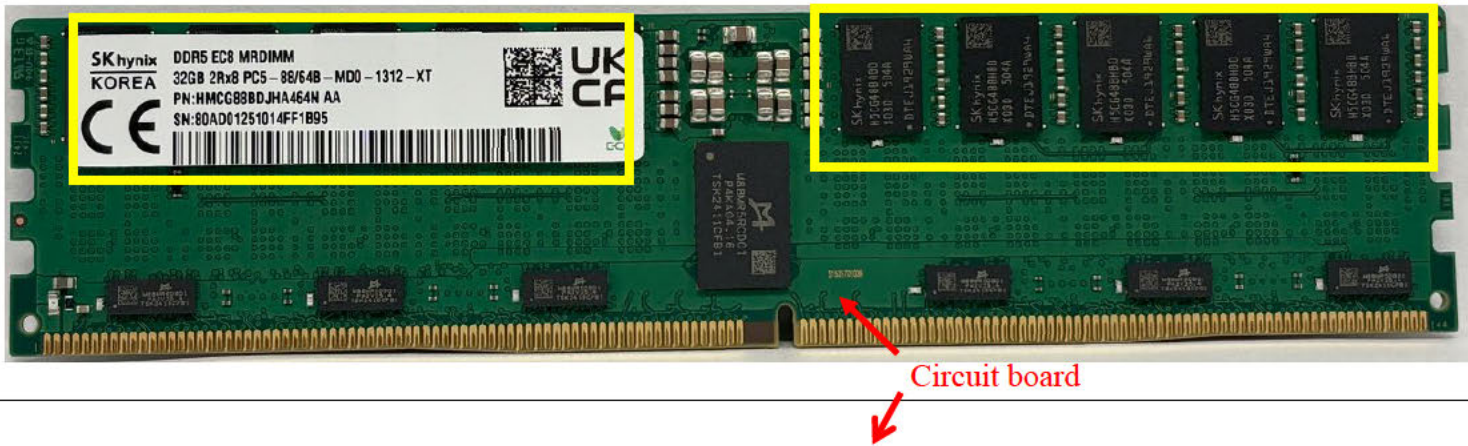
Montage's press releases similarly state that its "MRCD chip handles buffering and re-driving of address, command, clock, and control signals, while the *MDB chips manage data traffic between the memory controller and DRAM chips.*" See Ex. CC-66 (Montage Press Release) at 1:

The innovative MRCD and MDB chips are fundamental to MRDIMM operation, featuring a "1+10" buffer architecture where each MRDIMM module incorporates one MRCD with ten MDB chips. In this configuration, the MRCD chip handles buffering and re-driving of address, command, clock, and control signals, while the MDB chips manage data traffic between the memory controller and DRAM chips. This architecture, coupled with double data rate and time-division multiplexing technology, enables simultaneous operation of two memory ranks at standard rates, effectively doubling data throughput.

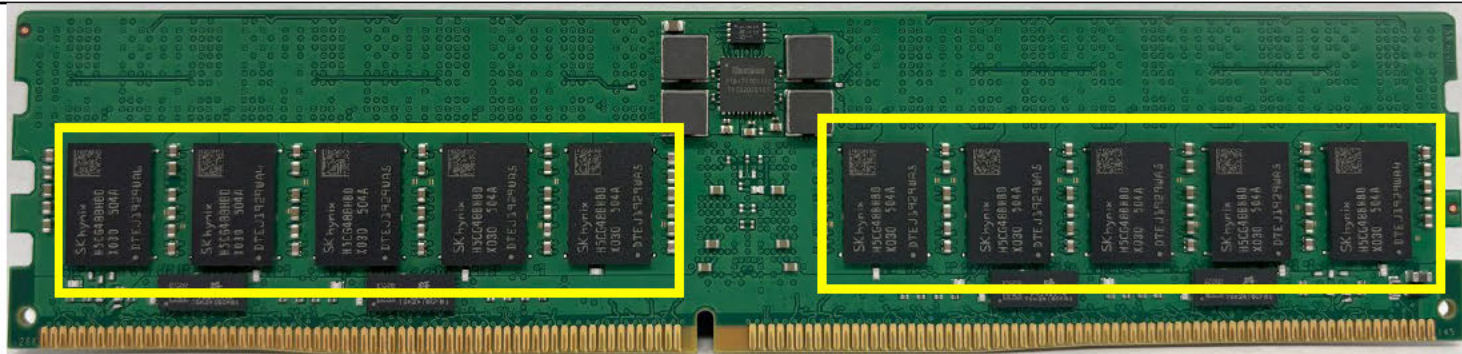


These descriptions by Montage, the DDR5 interface chip supplier for SK hynix's MRDIMMs, again demonstrate that an MRDIMM, like that of SK hynix's MRDIMM that incorporates Montage's MRCD and MDB chips, is designed to be accessible in a computer system by a system memory controller.

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	The communication of the data signals between the memory controller and the DDR5 SDRAMs on a SK hynix MRDIMM would be via a memory bus because a memory bus refers to the data pathway between a CPU and SDRAMs. <i>See, e.g.,</i> Ex. CC-67 (DevX Memory Bus Definition) at 2 (“Memory Bus” “refers to the data pathway between the central processing unit (CPU) and the computer’s main memory (RAM). Essentially, it serves as the communication channel that enables these two critical components to exchange information, ensuring the efficient functioning of the computer.”).
[1.b] [i] memory devices mounted on a circuit board, [ii] the memory devices having address and control ports and data ports;	[i] Each SK hynix MRDIMM includes memory devices mounted on a circuit board (also called a module board or a printed circuit board or PCB). As shown below, there are 20 total DRAM packages (highlighted in yellow boxes) mounted on a circuit board (the green PCB) with 10 on the back and 10 on the front, with each side of the MRDIMM having one row of 10 DRAMs. 

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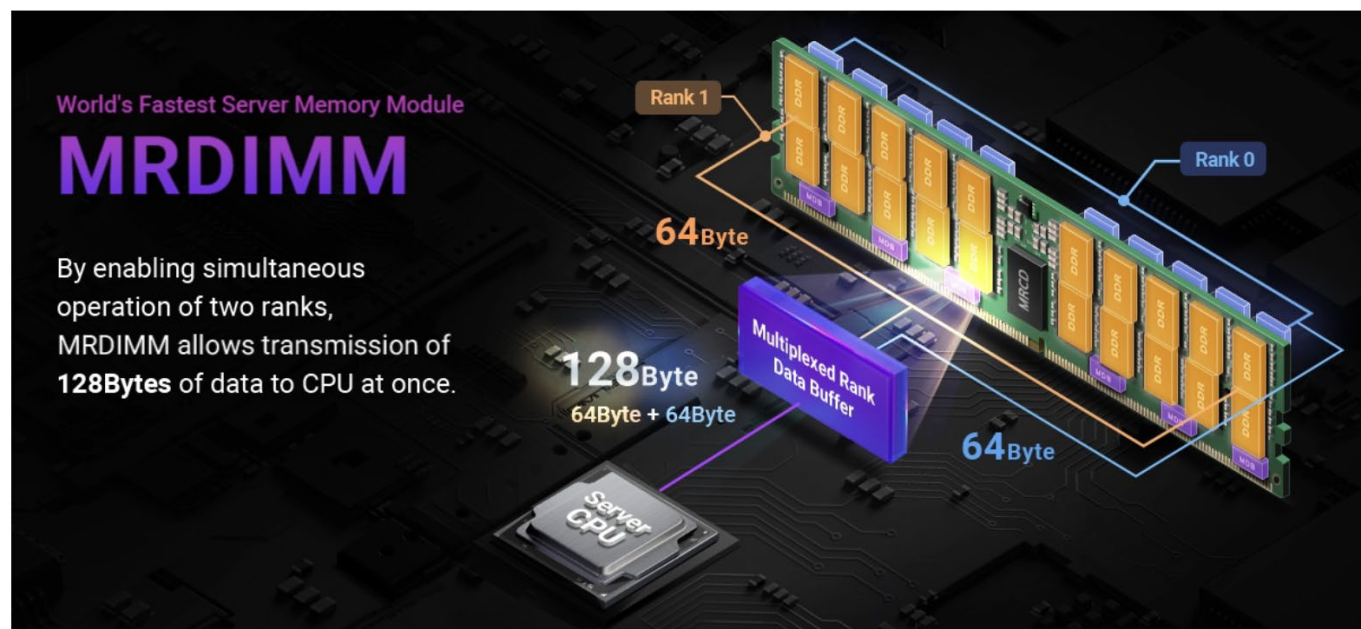
Ex. CC-36 (Product Images) at 13, 16 (annotated).

See, e.g., Ex. CC-42 (SK hynix MRDIMM) at 1–2 (showing multiple devices mounted on a PCB):

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How MRDIMM Works

By facilitating simultaneous operation of two ranks, MRDIMM enables the transmission of 128 bytes of data to the CPU at once, compared to the 64 bytes typically fetched by conventional DRAM modules. This increase in the amount of data sent to the CPU each time supports twice the speed of a single DRAM.

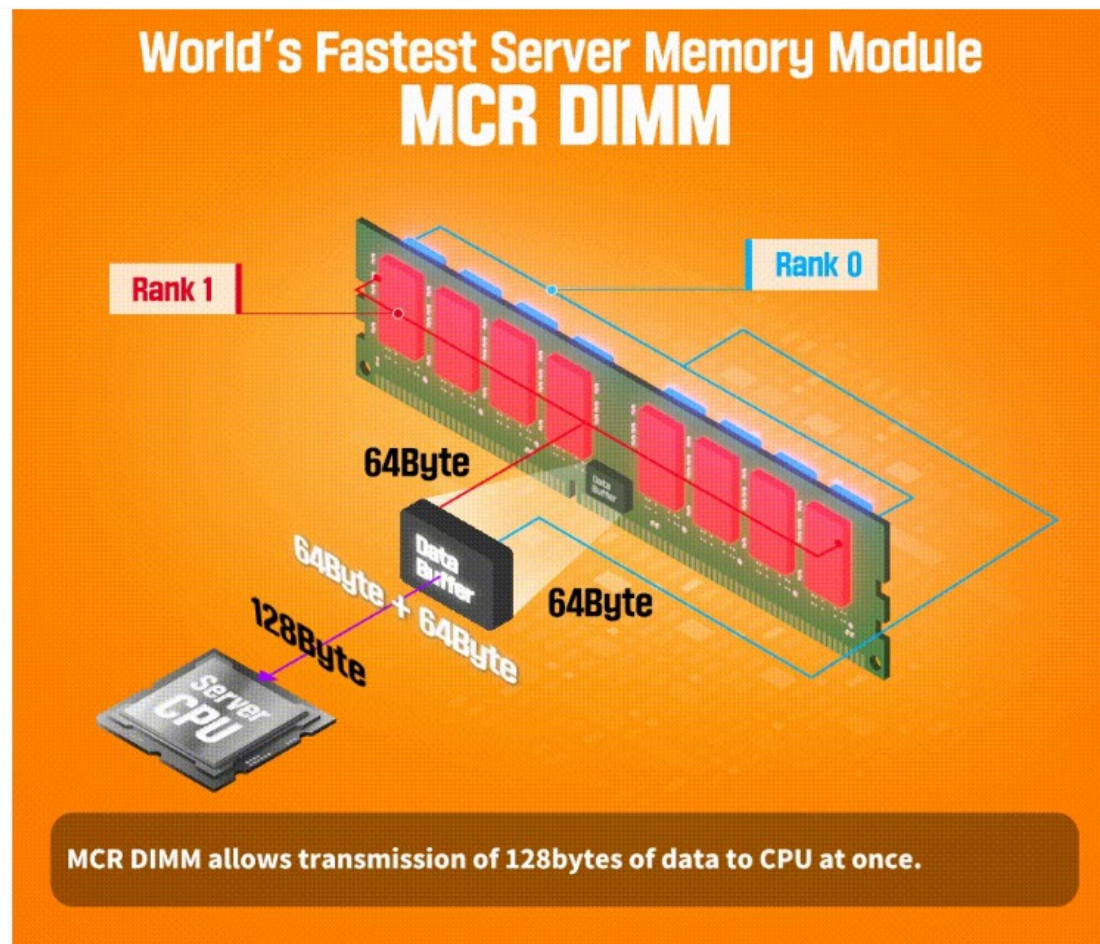


See Ex. CC-64 (SK hynix Newsroom Interview) at 2–3:

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SK hynix Pushes Technological Limits to Lead the Server DRAM Market

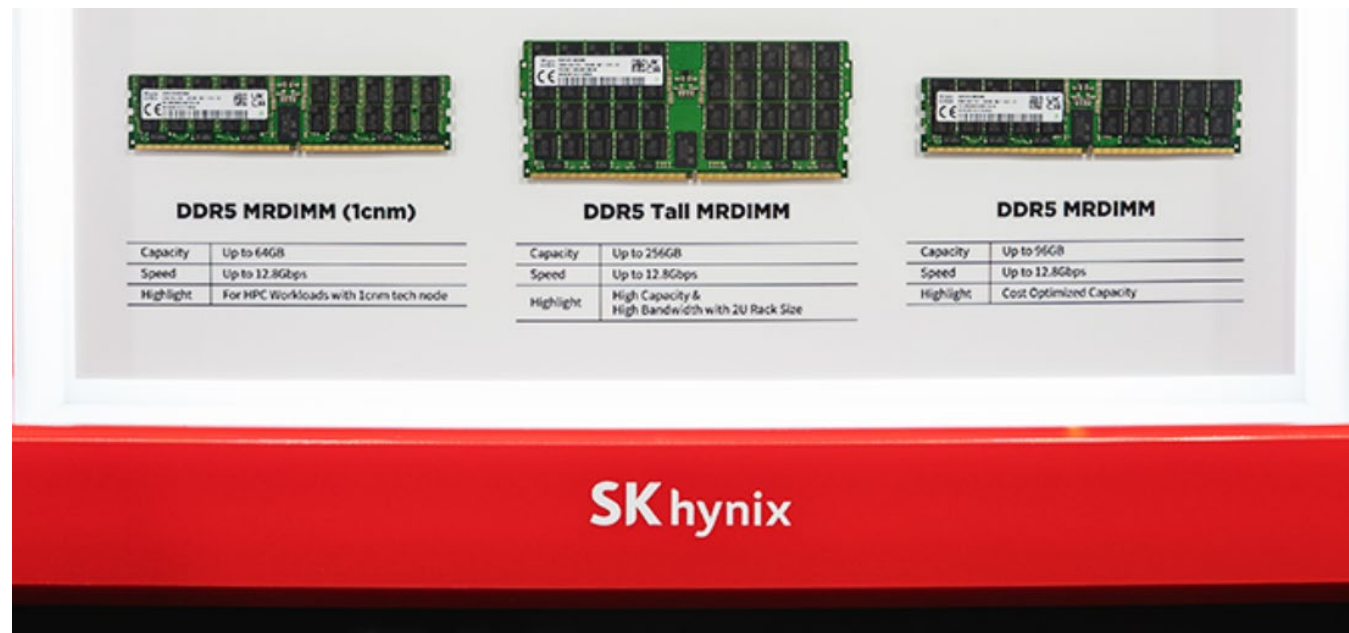
The MCR DIMM represents a leap forward in performance, typically unseen in the DRAM sector. It is common to see an operating speed increase of about 800 Mbps (Megabits per second) when developing a new generation of the DDR5 DRAM. For example, if the DRAM offered an operating speed of 4,800 Mbps, the next generation is expected to have a speed of approximately 5,600 Mbps. However, SK hynix made yet another breakthrough by developing the MCR DIMM that boasts an operating speed of over 8,000 Mbps.



▲ Figure 3. The operation structure of the world's fastest server DRAM, the MCR DIMM.

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Ex. CC-68 (SK hynix TSMC 2025 Technology Symposium) at 2 (excerpted):



DDR5 RDIMM and MRDIMM products on display, including those based on the 1c node

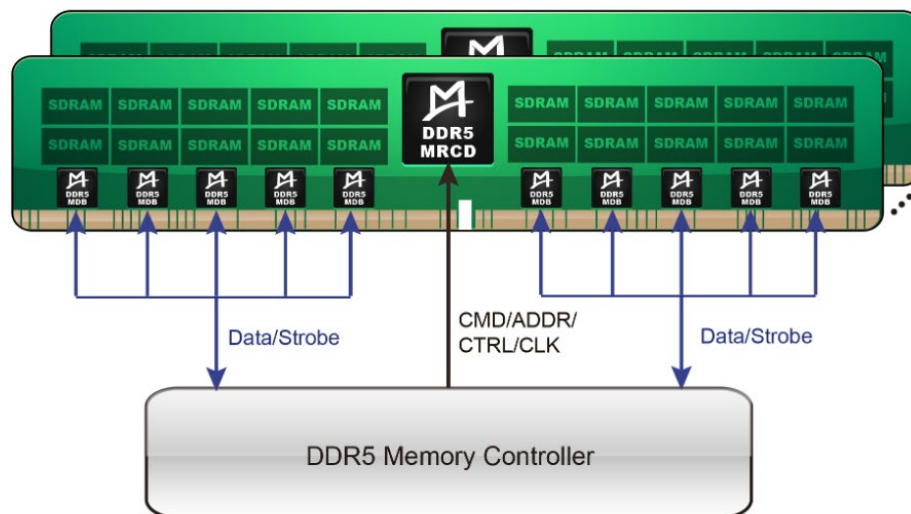
[ii] The memory devices have address and control ports and data ports. For example, the MRCD01 chip from Montage¹ is designed to “buffer[] and re-drive[] the address, command, clock and control signals from the memory controller.” Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer). These signals are to be driven to at least the memory devices in order to perform “efficient memory management operations.” *Id.* This implies that the memory devices have address and control ports, which are used to receive the address/control signals re-driven from the MRCD.

¹ The MRCD01 chip is an exemplary memory device. Other generations (e.g., 02) would be expected to have the same (or substantially the same) functionality and architectural configuration in relevant part.

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General Description

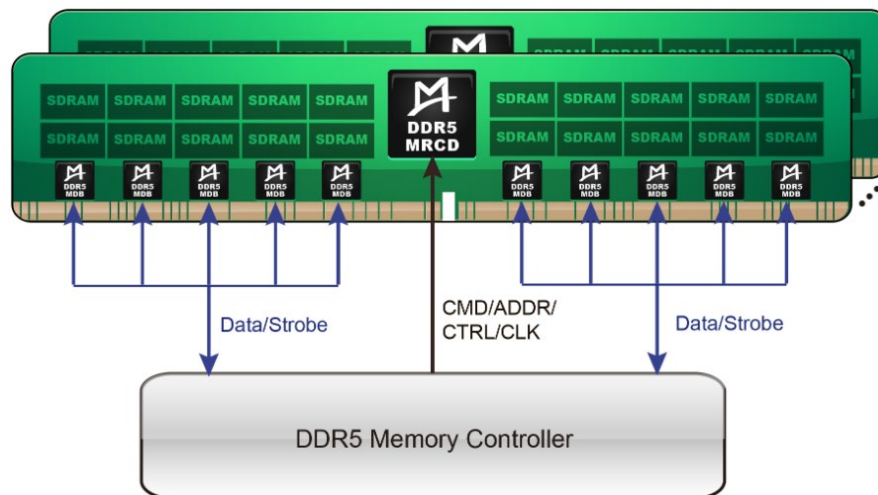
The M88MR5RCD01 is a first-generation DDR5 Multiplexed Rank Registering Clock Driver (MRCD) chip that supports data rates of up to 8800MT/s. This chip buffers and re-drives the address, command, clock, and control signals from the memory controller. Compared to traditional DDR5 RCD chips, the MRCD can generate four independent chip-select signals at standard rates, enabling more flexible and efficient memory management operations. Working in conjunction with Multiplexed Rank Data Buffer (MDB) chips, the MRCD chip is applied in next-generation DDR5 MRDIMM / MCRDIMM memory modules, effectively doubling memory system bandwidth to meet the urgent demand for memory throughput in applications such as deep learning and large-scale data analysis.



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	Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1–2 (showing command/address/control/clock signals being driven to MRCD). • Two sub-channels, each divided into two pseudo-channels to increase the total bandwidth of the host system • Two pseudo-channels splitting the incoming CA inputs and generating separate CA outputs Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1 (“sub-channels” and “pseudo-channels” imply memory devices, and the separately generated CA outputs are sent to memory devices). As another example, Montage’s MDB states the “MDB chip buffers and re-drives data signals from the memory controller or DRAM devices” and it “efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal.” Ex. CC-22 (Montage M88MR5DB01) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1. This implies that memory devices or DDR5 SDRAMs have data ports for receiving and outputting data signals. General Description The M88MR5DB01 is a Gen1 DDR5 Multiplexed Data Buffer (MDB) supporting data rates up to 8800MT/s. The MDB chip buffers and re-drives data signals from the memory controller or DRAM devices. It incorporates a dual-nibble host interface that operates at twice the speed of the DRAM interface. The DRAM side features four nibbles, with two allocated to each pseudo-channel. The MDB efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal. Designed to work in conjunction with the Multiplexed Registering Clock Driver (MRCD), the MDB plays a crucial role in DDR5 MRDIMMs / MCRDIMMs. Each MRDIMM / MCRDIMM is equipped with one MRCD and ten MDBs. This configuration enables the MRDIMM / MCRDIMM module to concurrently access two DRAM memory ranks through the MDB chips, resulting in a doubling of bandwidth to address the demands of high-load applications.
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Ex. CC-22 (Montage M88MR5DB01) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1–2.

Hence, the memory devices have control/address ports and data ports.

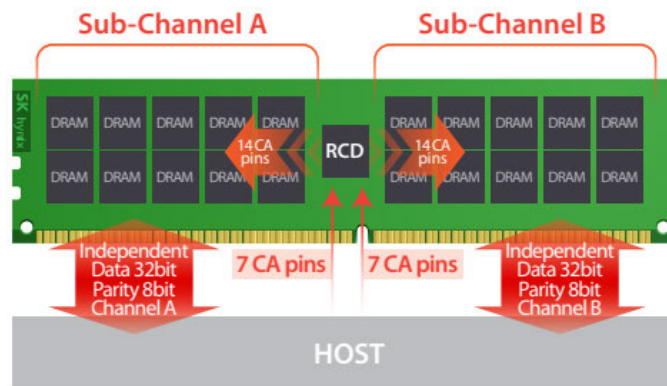
Additionally, it is expected that the same type of SDRAMs would be used on MRDIMMs and RDIMMs. Ex. CC-13 (JEDEC DDR5 MRDIMM Plan) (stating that one planned feature for MRDIMMs is “[u]tilization of standard DDR5 DIMM components including DRAM, DIMM Form Factor & Pinout, SPD, PMIC, and TS for ease of adoption”).

In SK hynix’s DDR5 RDIMM, each of the SDRAM units includes inputs for data signals and address/control signals. *See, e.g.*, Ex. CC-40 (SK hynix A New Standard) at 5:

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The latest DDR5 DRAM standard turbo-boosts RDIMM module performance and technology for the era of AI and Big Data. SK hynix expects its DDR5 RDIMMs to reach over 8,000Mbps in data rates with research into continued speed enhancements.

Features



| New Sub-channel DRAM Architecture

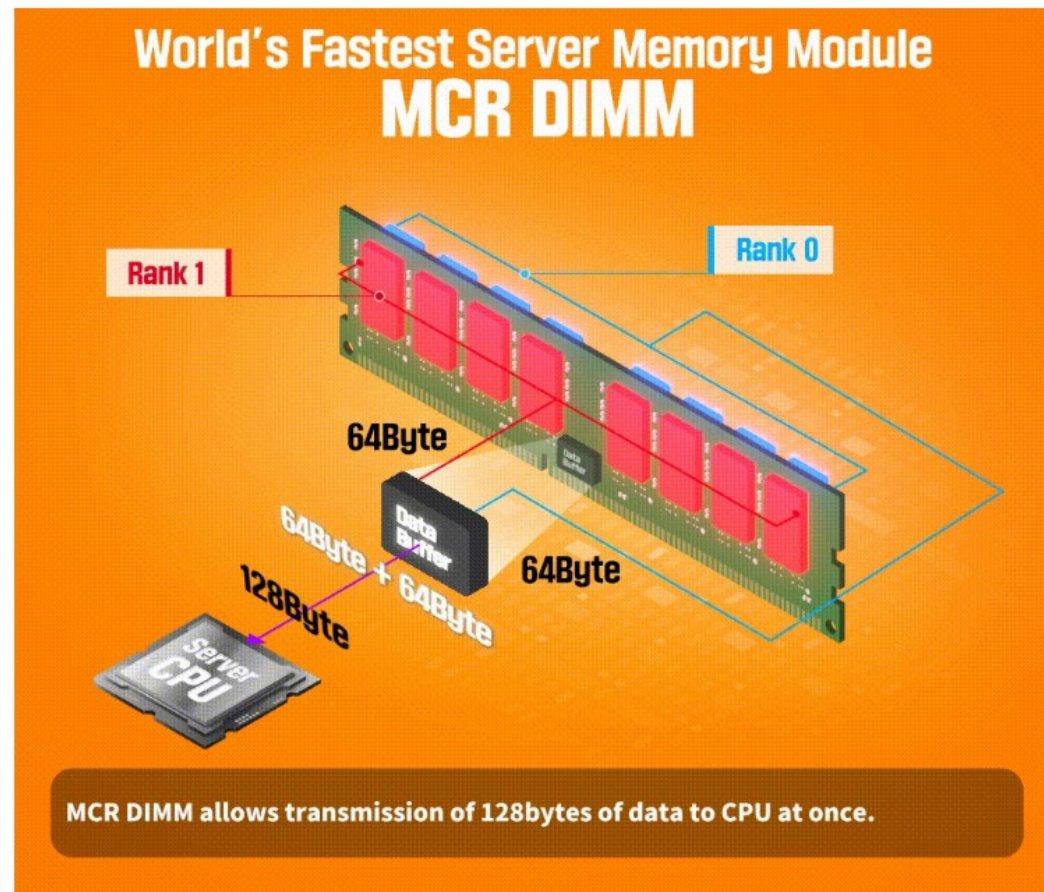
The dual sub-channel structure on DDR5 RDIMM allows for an increase in CA pin count, with seven CA buses per sub-channel running at double data rate (DDR) from host to RCD and 14 on a single data rate (SDR) interface from RCD to DRAM, effectively doubling bandwidth over a single DDR4 channel.

See also Ex. CC-64 (SK hynix Newsroom Interview) at 2–3:

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SK hynix Pushes Technological Limits to Lead the Server DRAM Market

The MCR DIMM represents a leap forward in performance, typically unseen in the DRAM sector. It is common to see an operating speed increase of about 800 Mbps (Megabits per second) when developing a new generation of the DDR5 DRAM. For example, if the DRAM offered an operating speed of 4,800 Mbps, the next generation is expected to have a speed of approximately 5,600 Mbps. However, SK hynix made yet another breakthrough by developing the MCR DIMM that boasts an operating speed of over 8,000 Mbps.



▲ Figure 3. The operation structure of the world's fastest server DRAM, the MCR DIMM.

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See also, e.g., Ex. CC-58 (Kim 2020) at 167 (This is an article by SK hynix, showing C/A and DQ (data) ports for C/A and DQ signaling.).

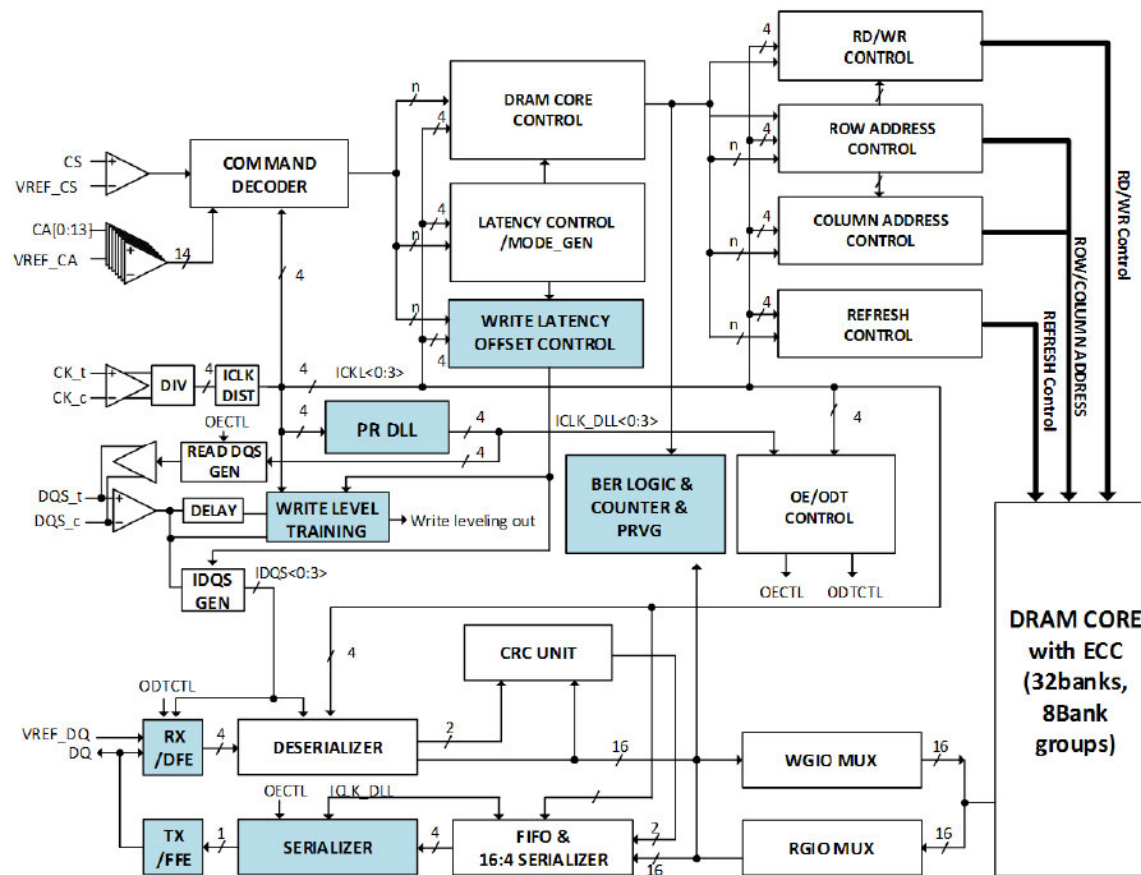


Fig. 2. DDR5 functional block diagram.

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Ex. CC-58 (Kim 2020) at 168, Fig. 2.

See also, e.g., Ex. CC-05 (JESD79-5C.01_v1.31) at Section 2.4 (“The definition of the signals (DQ, DQS, CA, CK) and the corresponding ball/signal assignments show that the memory devices have address and control ports (CA pins), as well as data ports (DQ/DQS pins).”):

2.4 DDR5 SDRAM x4/8 Ballout Using MO-210

Table 1 — DDR5 SDRAM x4/8 Ballout Using MO-210

AN	1	2	3	4	5	6	7	8	9	10	11
AL		1	2	3	4	5	6	7	8	9	
A	DNU	LBDQ	VSS	VPP				ZQ ⁷	VSS	LBDQS	DNU
B		VDD	VDDQ	DQ2				DQ3	VDDQ	VDD	
C		VSS	DQ0	DQS _t				DM _n ⁴ , TDQS _t ²	DQ1	VSS	
D		VDDQ	VSS	DQS _c				TDQS _c ³	VSS	VDDQ	
E		VDD	DQ4 ¹	DQ6 ¹				DQ7 ¹	DQ5 ¹	VDD	
F		VSS	VDDQ	VSS				VSS	VDDQ	VSS	
G		CA_ODT	MIR	VDD				CK _t	VDDQ	TEN ⁵ (CS1 _n ⁵)	
H		ALERT _n	VSS	CS _n				CK _c	VSS	VDD	
J		VDDQ	CA4	CA0				CA1	CA5	VDDQ	
K		VDD	CA6	CA2				CA3	CA7	VDD	
L		VDDQ	VSS	CA8				CA9	VSS	VDDQ	
M		CAI, ZQ1 ⁸	CA10	CA12				CA13	CA11	RESET _n	
N	DNU	VDD	VSS	VDD				VPP	VSS	VDD	DNU

NOTE 1 DQ4-DQ7 are higher order DQ pins and are not connected for the x4 configuration.

NOTE 2 TDQS_t is not valid for the x4 configuration

NOTE 3 TDQS_c is not available for the x4 configuration

NOTE 4 DM_n not valid for the x4 configuration

NOTE 5 For single die package (SDP) or 3DS, this pin is TEN; CS1_n is not connected, and is used as Test Mode Enable. For dual die package (DDP), this pin is CS1_n; and TEN is disconnected, and is used as CS1_n.

NOTE 6 The electrical parameters for DDP package are TBD and will be covered from future ballouts.

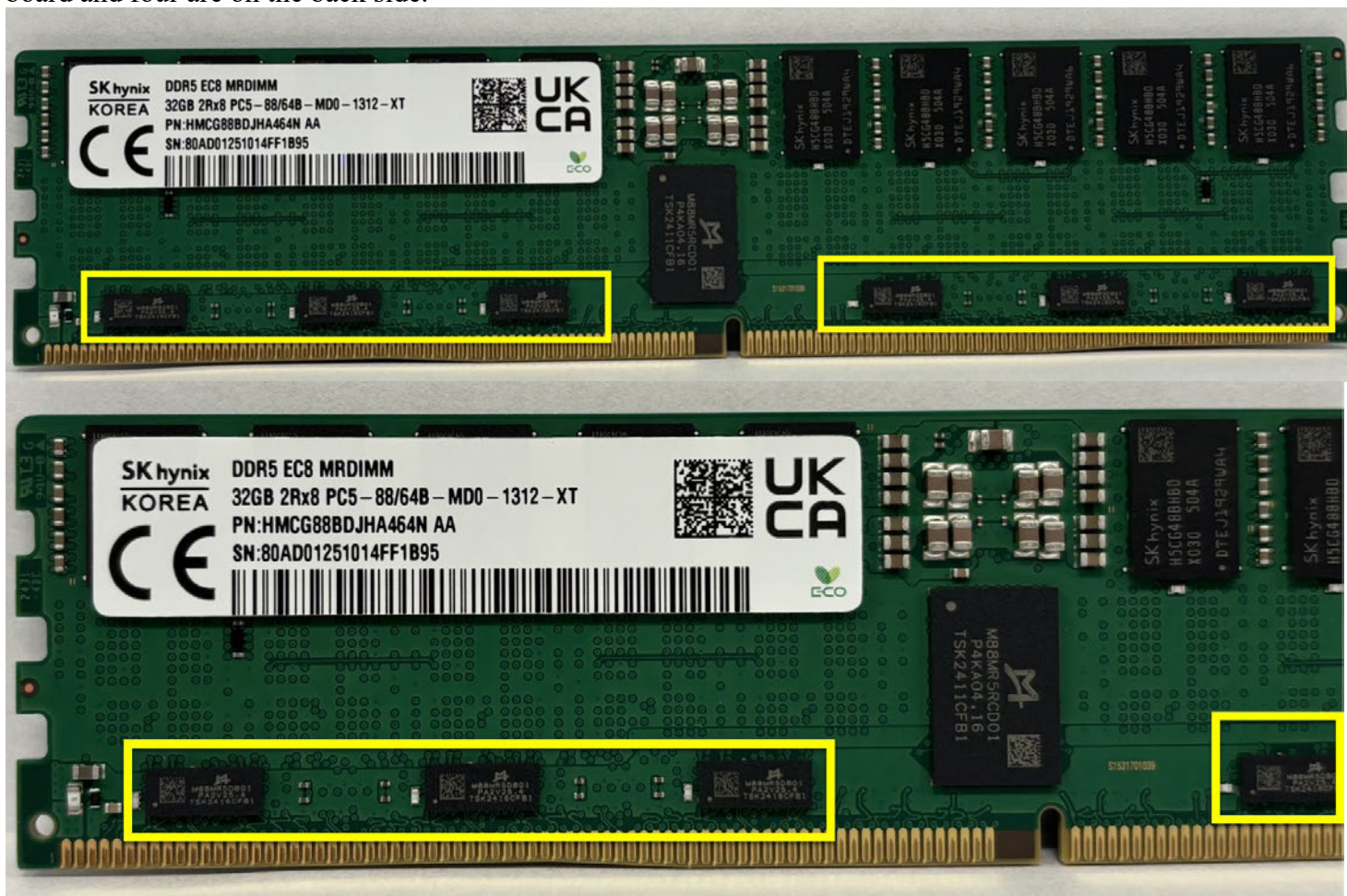
NOTE 7 For dual die package (DDP), ZQ is connected to bottom die (Rank 0).

NOTE 8 For single die package (SDP) or 3DS, CAI is used to indicate Command Address Inversion (ZQ1 is not applicable). However, for dual die package (DDP), CAI die pads are grounded inside the package, and ZQ1 pin is connected to the top die ZQ pad (Rank 1).

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[1.c] [i] a data module mounted on the circuit board and [ii] coupled between the data ports of the memory devices and the system memory bus,

[i] SK hynix's MRDIMMs each include a data module, such as a collection of ten MDBs (highlighted in yellow boxes), mounted on the circuit board, as illustrated below. As seen below, six of the MDBs are on the front side of the circuit board and four are on the back side.



Ex. CC-36 (Product Images) at 14, 15 (annotated).

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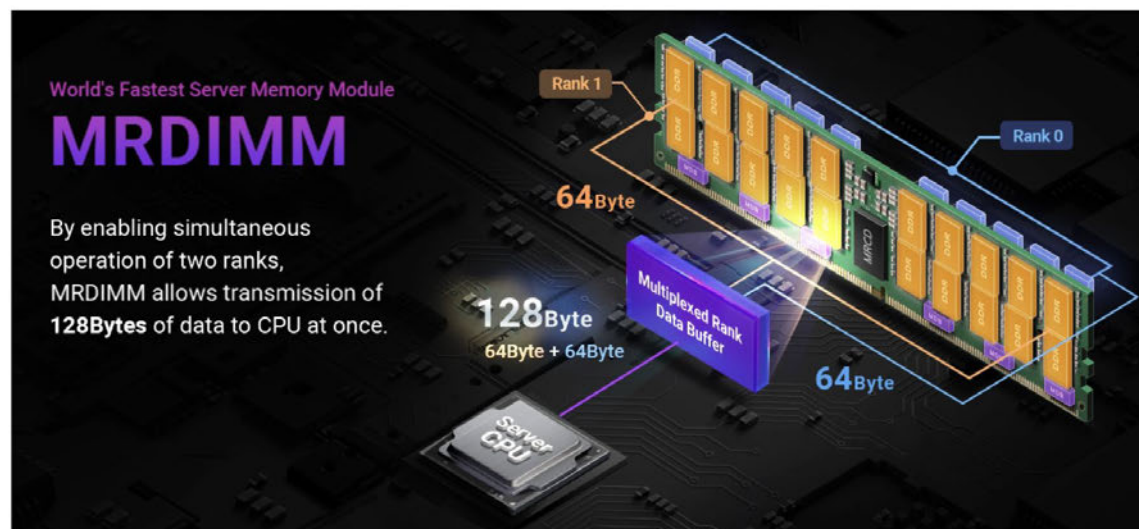
See also Ex. CC-22 (Montage M88MR5DB01); Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) (listing the product information for Montage's MDB).

[ii] The data module, i.e., the collection of MDBs, is coupled between the data ports of the memory devices and the system memory bus.

For example, as shown on SK hynix's website (Ex. CC-42 (SK hynix MRDIMM) at 1–2)), the multiplexed rank data buffers are coupled between the SDRAM devices and the CPU for data transmission. This indicates that the MDBs are coupled between the data ports of the SDRAM devices and the data connectors of the memory bus.

How MRDIMM Works

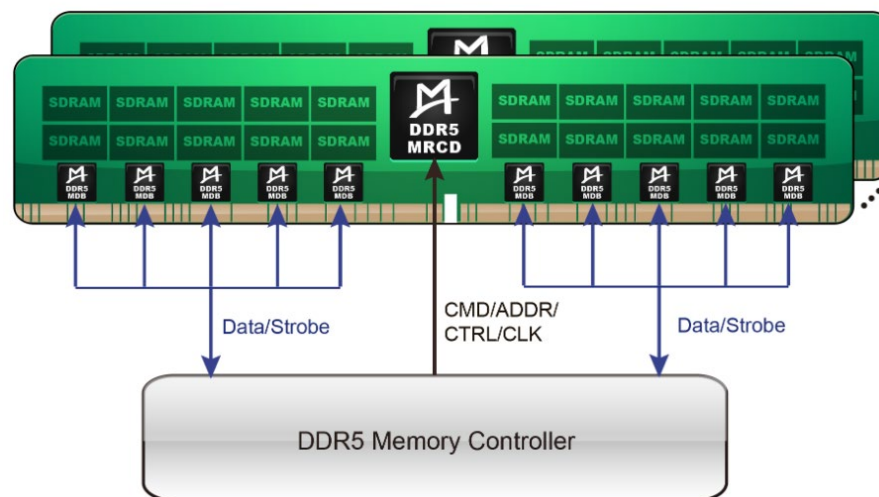
By facilitating simultaneous operation of two ranks, MRDIMM enables the transmission of 128 bytes of data to the CPU at once, compared to the 64 bytes typically fetched by conventional DRAM modules. This increase in the amount of data sent to the CPU each time supports twice the speed of a single DRAM.



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	Ex. CC-42 (SK hynix MRDIMM) at 1–2; <i>see</i> Ex. CC-64 (SK hynix Newsroom Interview). As already noted, Montage’s MDB states the “MDB chip buffers and re-drives data signals from the memory controller or DRAM devices” and it “efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal.” Ex. CC-22 (Montage M88MR5DB01) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1. Additionally, the MDB features “[d]ual nibbles on the host side,” “[f]our nibbles on the DRAM side, two per pseudo-channel,” and “[t]wo DRAM-side DQ signals multiplexed to one host-side DQ signal.” <i>Id.</i> This suggests that each data buffer is coupled between a respective set of memory devices and the memory controller, as illustrated on the Montage website: General Description The M88MR5DB01 is a Gen1 DDR5 Multiplexed Data Buffer (MDB) supporting data rates up to 8800MT/s. The MDB chip buffers and re-drives data signals from the memory controller or DRAM devices. It incorporates a dual-nibble host interface that operates at twice the speed of the DRAM interface. The DRAM side features four nibbles, with two allocated to each pseudo-channel. The MDB efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal. Designed to work in conjunction with the Multiplexed Registering Clock Driver (MRCD), the MDB plays a crucial role in DDR5 MRDIMMs / MCRDIMMs. Each MRDIMM / MCRDIMM is equipped with one MRCD and ten MDBs. This configuration enables the MRDIMM / MCRDIMM module to concurrently access two DRAM memory ranks through the MDB chips, resulting in a doubling of bandwidth to address the demands of high-load applications.
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Feature

- Speed up to 8800 MT/s
- Dual nibbles on the host side
- Four nibbles on the DRAM side, two per pseudo-channel
- Two DRAM-side DQ signals multiplexed to one host-side DQ signal
- One input-only control bus interface for connecting to a Multiplexed Rank Registering Clock Driver (MRCD)
- x4 and x8 DRAMs support
- Strobe and data training support
- DRAM periodic update support
- Multiple power-saving modes
- Transparent, continuous burst, loopback and pass-through modes support
- 1.1V VDD voltage
- Green package: 78-Ball FBGA

Ex. CC-22 (Montage M88MR5DB01) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1–2.

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[1.c] [iii] the data module including data handler logic elements; and

The data module on each of SK hynix's MRDIMMs includes data handler logic elements. For example, Montage states that its MDBs feature "[s]trobe and data training support" as well as circuit to multiplex two DRAM-side DQ signals to one host-side DQ signal. *See* Ex. CC-22 (Montage M88MR5DB01) at 1. This suggests that the data buffer has logic elements for data handling.

Furthermore, Montage states that it is the "spec owner of DDR5 . . . MDB . . . chips" and "have taken the lead in formulating international standards for DDR5 RCD, MDB, and CKD chips[.]" Ex. CC-62 (Montage HK Proof) at 163, 196. Thus, on information and belief,

See, e.g., Ex. CC-18C (MDB01 Spec. Rev. 0.7), §§ 8.8–8.8.2:

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[1.d] [i] a control module mounted on the circuit board and [ii] coupled to	[i] SK hynix's DDR5 MRDIMMs each contain a control module, such as the MRCD, mounted on the circuit board, as illustrated below:

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the [a] data module, [b] the address and control ports of the memory devices, and [c] the system memory bus; and



Ex. CC-36 (Product Images) at 13, 14 (annotated).

[ii.a] The control module (MRCD) is coupled to the data module. For example, one feature of Montage's MRCD is an "[i]ntegrated BCOM bus to control the multiplexed data buffers (MDB)." Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1; Ex. CC-22 (Montage M88MR5DB01). The MRCD is therefore coupled to the MDBs in the data module via the BCOM bus to control their operation. *See also, e.g.,* Ex. CC-22 (Montage M88MR5DB01) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1 (Montage's MDB features "[o]ne input-only control bus interface for connecting to a Multiplexed Rank Registering Clock Driver (MRCD)," which would be used to receive the control signals outputted by the MRCD).

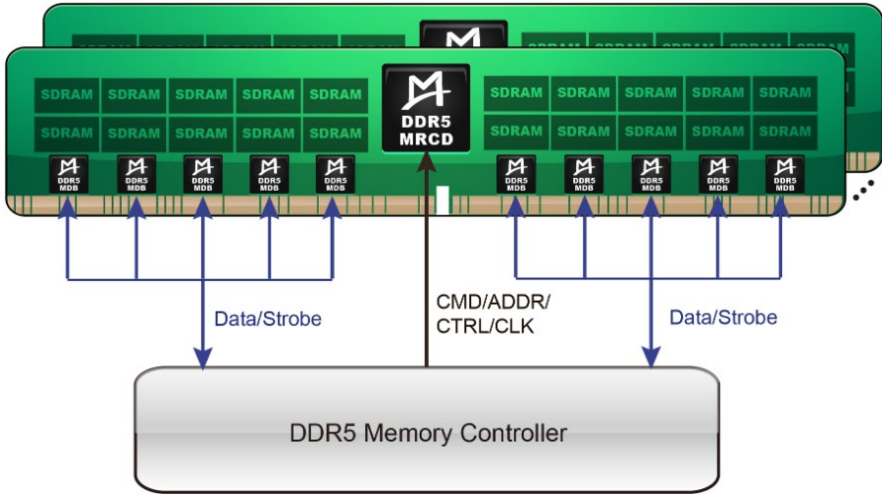
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	[ii.b] The MRCD is also coupled to the address and control ports of the memory devices. For example, as noted above, the MRCD01 chip from Montage is designed to “buffer[] and re-drive[] the address, command, clock, and control signals from the memory controller.” Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer). These signals are understood to be driven to at least the memory devices in order to perform “efficient memory management operations.” Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1. General Description The M88MR5RCD01 is a first-generation DDR5 Multiplexed Rank Registering Clock Driver (MRCD) chip that supports data rates of up to 8800MT/s. This chip buffers and re-drives the address, command, clock, and control signals from the memory controller. Compared to traditional DDR5 RCD chips, the MRCD can generate four independent chip-select signals at standard rates, enabling more flexible and efficient memory management operations. Working in conjunction with Multiplexed Rank Data Buffer (MDB) chips, the MRCD chip is applied in next-generation DDR5 MRDIMM / MCRDIMM memory modules, effectively doubling memory system bandwidth to meet the urgent demand for memory throughput in applications such as deep learning and large-scale data analysis. • Two sub-channels, each divided into two pseudo-channels to increase the total bandwidth of the host system • Two pseudo-channels splitting the incoming CA inputs and generating separate CA outputs <i>Id.</i> (“sub-channels” and “pseudo-channels” imply memory devices, and the separately generated CA outputs are sent to memory devices). <i>See also, e.g.,</i> Ex. CC-62 (Montage HK Proof) at 43, noting the receiving and output of command signals:
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	“MRCD” multiplexed rank registering clock driver, a specialized buffer chip used in memory modules to buffer and re-drive clock, address, and command signals efficiently across multiple memory ranks, improving signal integrity and enabling higher memory speeds and capacities Buffering and re-driving clock, address, and command signals “across multiple memory ranks” imply that the MRCD is coupled to address/control ports of the memory devices; otherwise, it won’t be able to drive the signals across multiple ranks of the memory devices. [ii.c] The MRCD is also coupled to the system memory controller, as noted by Montage. <i>See</i> Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) (“This [MRCD] chip buffers and re-drives the address, command, clock and control signals from the memory controller.”). General Description The M88MR5RCD01 is a first-generation DDR5 Multiplexed Rank Registering Clock Driver (MRCD) chip that supports data rates of up to 8800MT/s. This chip buffers and re-drives the address, command, clock, and control signals from the memory controller. Compared to traditional DDR5 RCD chips, the MRCD can generate four independent chip-select signals at standard rates, enabling more flexible and efficient memory management operations. Working in conjunction with Multiplexed Rank Data Buffer (MDB) chips, the MRCD chip is applied in next-generation DDR5 MRDIMM / MCRDIMM memory modules, effectively doubling memory system bandwidth to meet the urgent demand for memory throughput in applications such as deep learning and large-scale data analysis.
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	 Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1–2 (showing MRCD coupled to DDR5 Memory Controller).
[1.e] wherein the memory module is operable in any of a plurality of modes including a first mode and a second mode;	SK hynix's MRDIMMs are operable in any of a plurality of modes including a first mode and a second mode. For example, the MDB supports and is therefore operable in a normal read/write operation mode. Ex. CC-22 (Montage M88MR5DB01) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1 ("The M88MR5DB01 is a Gen1 DDR5 Multiplexed Data Buffer (MDB) supporting data rates up to 8800MT/s. The MDB chip buffers and re-drives data signals from the memory controller or DRAM devices. It incorporates a dual-nibble host interface that operates at twice the speed of the DRAM interface. The DRAM side features four nibbles, with two allocated to each pseudo-channel. The MDB efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal.").

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General Description

The M88MR5DB01 is a Gen1 DDR5 Multiplexed Data Buffer (MDB) supporting data rates up to 8800MT/s. The MDB chip buffers and re-drives data signals from the memory controller or DRAM devices. It incorporates a dual-nibble host interface that operates at twice the speed of the DRAM interface. The DRAM side features four nibbles, with two allocated to each pseudo-channel. The MDB efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal.

Designed to work in conjunction with the Multiplexed Registering Clock Driver (MRCD), the MDB plays a crucial role in DDR5 MRDIMMs / MCRDIMMs. Each MRDIMM / MCRDIMM is equipped with one MRCD and ten MDBs. This configuration enables the MRDIMM / MCRDIMM module to concurrently access two DRAM memory ranks through the MDB chips, resulting in a doubling of bandwidth to address the demands of high-load applications.

As another example, Montage's MDB supports "[s]trobe and data training," and "[t]ransparent, continuous burst, loopback and pass-through modes" as well as "[m]ultiple power-saving modes." Ex. CC-22 (Montage M88MR5DB01) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1.

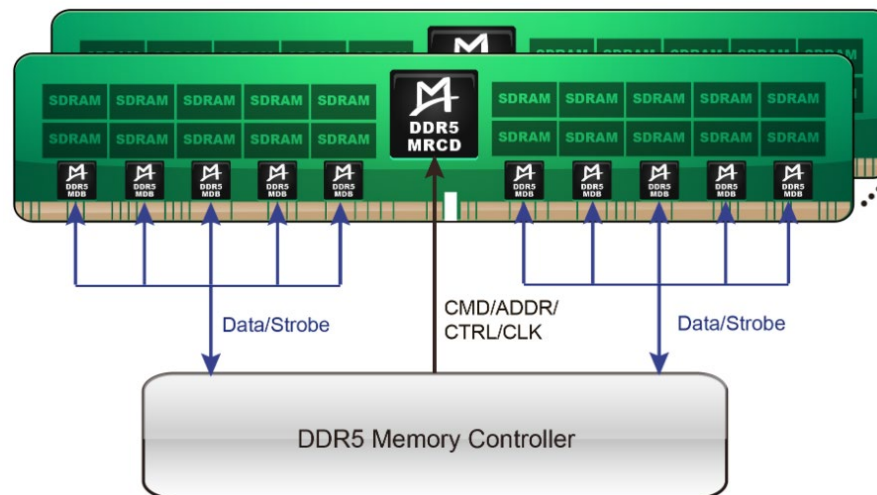
Feature

- Speed up to 8800 MT/s
- Dual nibbles on the host side
- Four nibbles on the DRAM side, two per pseudo-channel
- Two DRAM-side DQ signals multiplexed to one host-side DQ signal
- One input-only control bus interface for connecting to a Multiplexed Rank Registering Clock Driver (MRCD)
- x4 and x8 DRAMs support
- Strobe and data training support
- DRAM periodic update support
- Multiple power-saving modes
- Transparent, continuous burst, loopback and pass-through modes support
- 1.1V VDD voltage
- Green package: 78-Ball FBGA

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	Ex. CC-22 (Montage M88MR5DB01) at 1. An example of a second mode is a training mode.
[1.f] [i] wherein the control module in the first mode is configured to receive system address and control signals from the system memory controller and to output first memory address and control signals to the memory devices according to the system address and control signals,	In each of SK hynix’s DDR5 MRDIMMs, the control module, i.e., the MRCD, in the first mode is configured to receive system address and control signals from the system memory controller and to output first memory address and control signals to the memory devices according to the system address and control signals. For example, Montage describes its MRCD as handling “buffer[ing] and re-driv[ing] the address, command, clock, and control signals from the memory controller,” which means the MRCD is designed to receive the address/<i>command</i>/clock/control signals from the memory controller for otherwise it would not be able to buffer or re-drive those signals. <i>See</i> Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer). General Description The M88MR5RCD01 is a first-generation DDR5 Multiplexed Rank Registering Clock Driver (MRCD) chip that supports data rates of up to 8800MT/s. This chip buffers and re-drives the address, command, clock, and control signals from the memory controller. Compared to traditional DDR5 RCD chips, the MRCD can generate four independent chip-select signals at standard rates, enabling more flexible and efficient memory management operations. Working in conjunction with Multiplexed Rank Data Buffer (MDB) chips, the MRCD chip is applied in next-generation DDR5 MRDIMM / MCRDIMM memory modules, effectively doubling memory system bandwidth to meet the urgent demand for memory throughput in applications such as deep learning and large-scale data analysis.

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Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1–2 (showing MRCD coupled to DDR5 Memory Controller).

As also explained, the control module (MRCD) is also configured, in the first mode (normal read/write), to output first memory address and control signals to the memory devices according to the system address and control signals. For example, the Montage describes its MRCD as including the following features, where the incoming control/address inputs are split and separate control/address outputs are then generated based on the split incoming C/A signals:

- Two sub-channels, each divided into two pseudo-channels to increase the total bandwidth of the host system
- Two pseudo-channels splitting the incoming CA inputs and generating separate CA outputs

Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer).

Additionally, Ex. CC-62 (Montage HK Proof) at 43 defines MRCD as:

[REDACTED]

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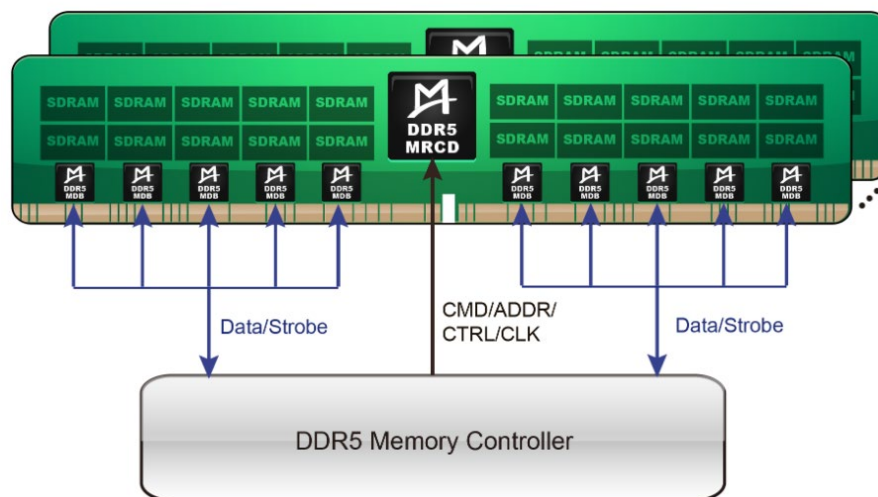
	“MRCD” multiplexed rank registering clock driver, a specialized buffer chip used in memory modules to buffer and re-drive clock, address, and command signals efficiently across multiple memory ranks, improving signal integrity and enabling higher memory speeds and capacities The command/control/address signals re-driven across the multiple memory ranks of memory devices are those received and buffered from the system memory controller.
[1.f] [ii] and the data module in the first mode is configured to propagate one or more first data signals between the memory devices and the system memory controller,	In the first mode (normal read/write), the data module (the collection of ten MDBs) is configured to propagate one or more first data signals between the memory devices and the system memory controller. As already noted, Montage’s MDB chips each “buffer[] and re-drive[] data signals from the memory controller or DRAM devices” and “efficiently multiplex[] two DRAM-side DQ signals into one host-side DQ signal.” Ex. CC-22 (Montage M88MR5DB01) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1. This suggests that the data buffers are configured to propagate one or more data signals between the memory devices and the system memory controller:

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General Description

The M88MR5DB01 is a Gen1 DDR5 Multiplexed Data Buffer (MDB) supporting data rates up to 8800MT/s. The MDB chip buffers and re-drives data signals from the memory controller or DRAM devices. It incorporates a dual-nibble host interface that operates at twice the speed of the DRAM interface. The DRAM side features four nibbles, with two allocated to each pseudo-channel. The MDB efficiently multiplexes two DRAM-side DQ signals into one host-side DQ signal.

Designed to work in conjunction with the Multiplexed Registering Clock Driver (MRCD), the MDB plays a crucial role in DDR5 MRDIMMs / MCRDIMMs. Each MRDIMM / MCRDIMM is equipped with one MRCD and ten MDBs. This configuration enables the MRDIMM / MCRDIMM module to concurrently access two DRAM memory ranks through the MDB chips, resulting in a doubling of bandwidth to address the demands of high-load applications.



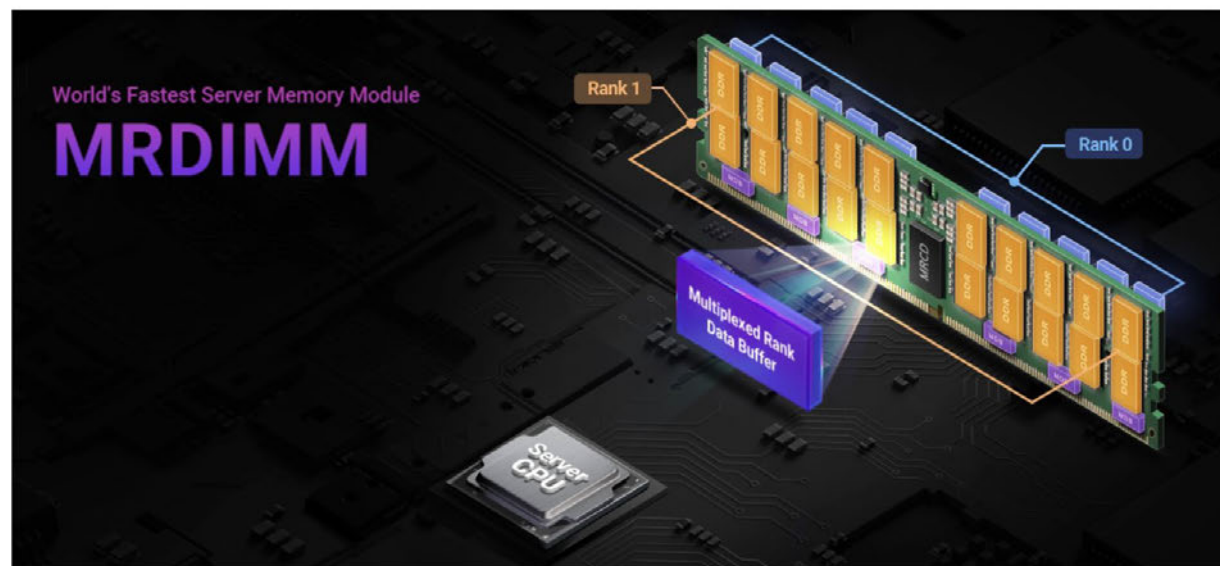
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	Feature • Speed up to 8800 MT/s • Dual nibbles on the host side • Four nibbles on the DRAM side, two per pseudo-channel • Two DRAM-side DQ signals multiplexed to one host-side DQ signal • One input-only control bus interface for connecting to a Multiplexed Rank Registering Clock Driver (MRCD) • x4 and x8 DRAMs support • Strobe and data training support • DRAM periodic update support • Multiple power-saving modes • Transparent, continuous burst, loopback and pass-through modes support • 1.1V VDD voltage • Green package: 78-Ball FBGA Ex. CC-22 (Montage M88MR5DB01) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) at 1–2. <i>See also</i> Ex. CC-42 (SK hynix MRDIMM) at 1–2 (showing data being propagated between the memory devices and the memory controller on CPU):
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How MRDIMM Works

By facilitating simultaneous operation of two ranks, MRDIMM enables the transmission of 128 bytes of data to the CPU at once, compared to the 64 bytes typically fetched by conventional DRAM modules. This increase in the amount of data sent to the CPU each time supports twice the speed of a single DRAM.



[1.f] [iii] the one or more first data signals being transmitted or received by at least a portion of the memory devices in

In each of SK hynix's MRDIMMs, the one or more first data signals are being transmitted or received by at least a portion of the memory devices in response to the first memory address and control signals. For example, in DDR5, commands for memory operations are defined by a combination of C/A and CS signals. *See* Ex. CC-46 (DDR5 Command Truth Table) (also included at Ex. CC-69 ((LP)DDR5 Memory Interface Presentation) at 11) (showing that commands for Activate, Read, and Write are composed of C/A[0:13] signals):

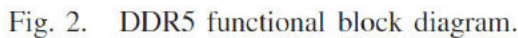
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response to the first memory address and control signals; and

Function	Abbreviation	CS_n	CA Pins													NOTES	
			CA0	CA1	CA2	CA3	CA4	CA5	CA6	CA7	CA8	CA9	CA10	CA11	CA12		CA13
Activate	ACT	L	L	L	R0	R1	R2	R3	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DDPIO	11,17,20,21,23,24
		H	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13	R14	R15	R16	CID3/R17	
RFU	RFU	L	H	L	L	L	L	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
RFU	RFU	L	H	L	L	L	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Write Pattern	WRP	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DDPIO	11,15,18,19,20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	H	V	CID3	
Write Pattern w/ Auto Precharge	WRPA	L	H	L	L	H	L	H	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DDPIO	11,15,18,19,20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	H	V	CID3	
RFU	RFU	L	H	L	L	H	H	V	V	V	V	V	V	V	V	V	
		H	V	V	V	V	V	V	V	V	V	V	V	V	V	V	
Mode Register Write	MRW	L	H	L	H	L	L	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V/DDPIO	8,11,13,20,22,23,24
		H	OP0	OP1	OP2	OP3	OP4	OP5	OP6	OP7	V	V	CW	V	V	V	
Mode Register Read	MRR	L	H	L	H	L	H	MRA0	MRA1	MRA2	MRA3	MRA4	MRA5	MRA6	MRA7	V/DDPIO	8,13,20,22,23,24,25,26
		H	L	L	V	V	V	V	V	V	V	V	CW	V	V	V	
Write	WR	L	H	L	H	H	L	BL=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DDPIO	8,12,15,19,20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	H	WR, Partial=1	V	CID3	
Write w/Auto Precharge	WRA	L	H	L	H	H	L	BL=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DDPIO	8,12,15,19,20,21,23,24
		H	V	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	WR, Partial=1	V	CID3	
Read	RD	L	H	L	H	H	H	BL=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DDPIO	8,15,19,20,21,23,24
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	H	V	V	CID3	
Read w/Auto Precharge	RDA	L	H	L	H	H	H	BL=L	BA0	BA1	BG0	BG1	BG2	CID0	CID1	CID2/DDPIO	8,15,19,20,21,23,24
		H	C2	C3	C4	C5	C6	C7	C8	C9	C10	V	AP=L	V	V	CID3	

The read and write operation and the associated transmission and receipt of the one or more data signals are in response to the first memory control and address signals received by the memory devices. Depending on the value of the chip select signal, a portion of or all of the memory devices may participate in the data transmission and receipt.

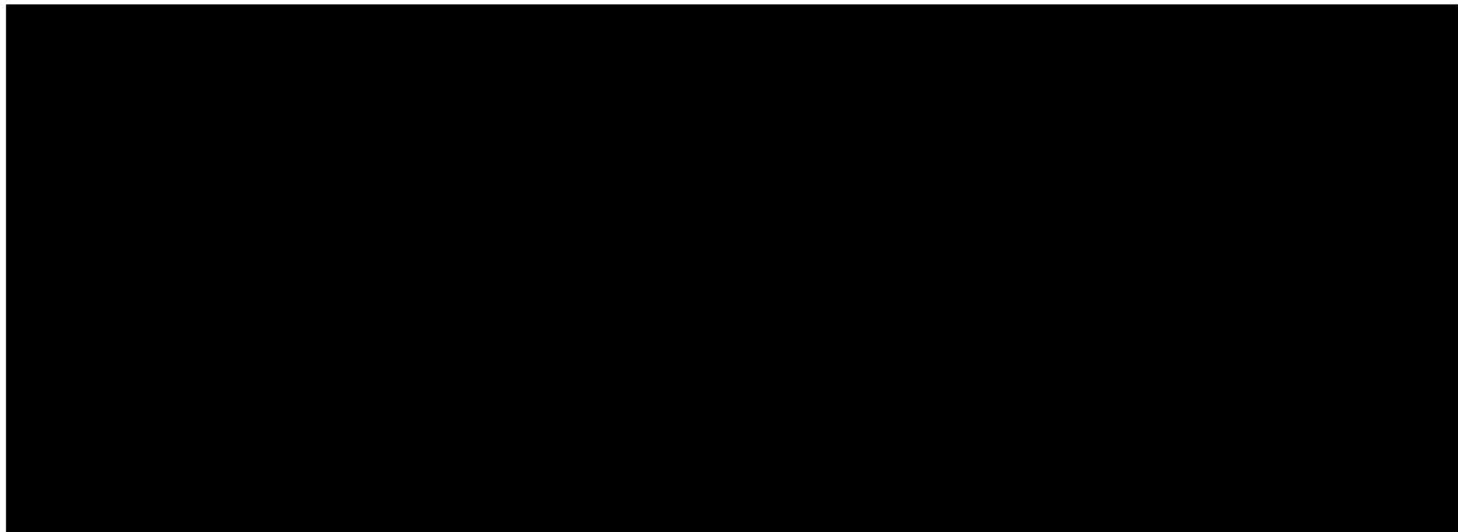
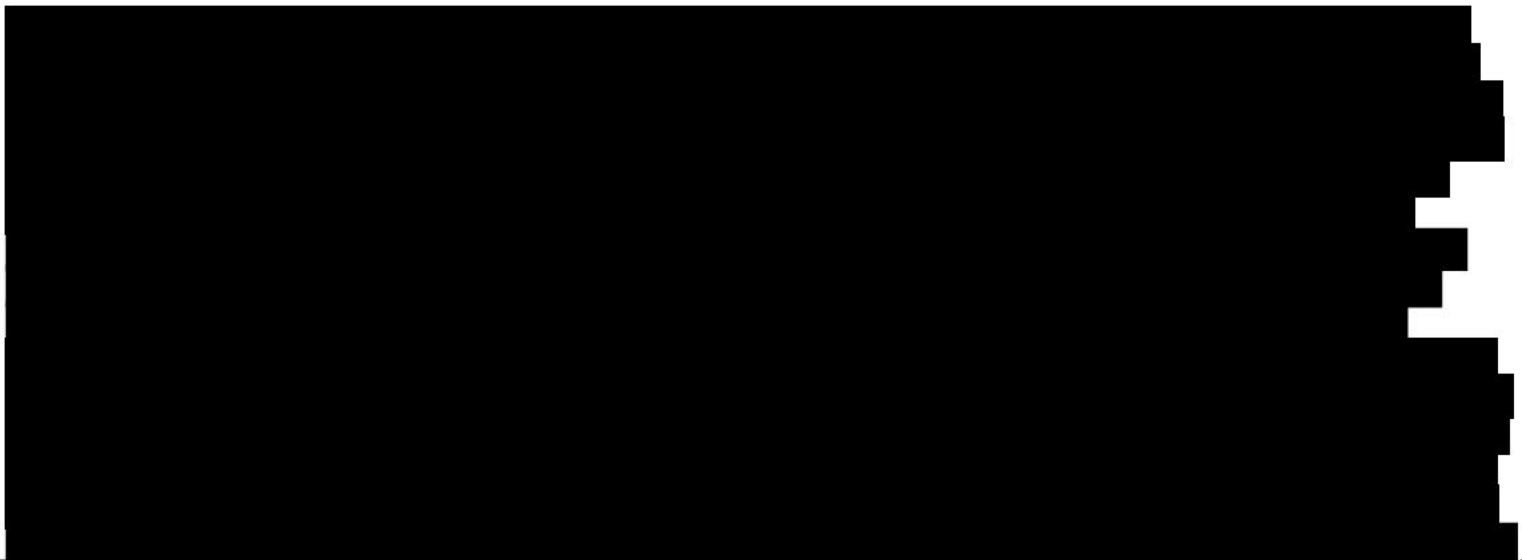
Ex. CC-58 (Kim 2020) at 168, Fig. 2 (annotated) (a SK hynix article illustrating basic functional blocks of a DDR5 SDRAM, showing that read/write control is in response to CS and CA signal inputs to the SDRAM device which signals collectively define commands for memory operations):



Each of SK hynix's MRDIMMs is expected to meet this claim element consistent with the JEDEC standard. *See, e.g.*, Ex. CC-05 (JESD79-5C.01 v1.31) at Section 4.1 (Command Truth Table).

In each of SK hynix’s MRDIMMs, the control module (MRCD) in the second mode (training mode) is configured to output second memory address and control signals to the address and control ports of the memory devices. For example, Montage, being the “spec owner of DDR5 . . . MDB . . . chips” (Ex. CC-62 (Montage HK Proof) at 163), likely implements features in MDB Specification. Montage supports data and strobe training. Ex. CC-22 (Montage

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second memory address and control signals to the address and control ports of the memory devices,	M88MR5DB01); Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer). And training includes training between DRAM and MDBs. <i>See</i> Ex. CC-18C (MDB01 Spec. Rev. 0.7) at 108, Table 38:  
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See also Ex. CC-58 (Kim 2020) at 168, Fig. 2 (annotated) (a SK hynix article illustrating basic functional blocks of a DDR5 SDRAM, showing that read/write control is in response to CS and CA signal inputs to the SDRAM device which signals collectively define commands for memory operations):

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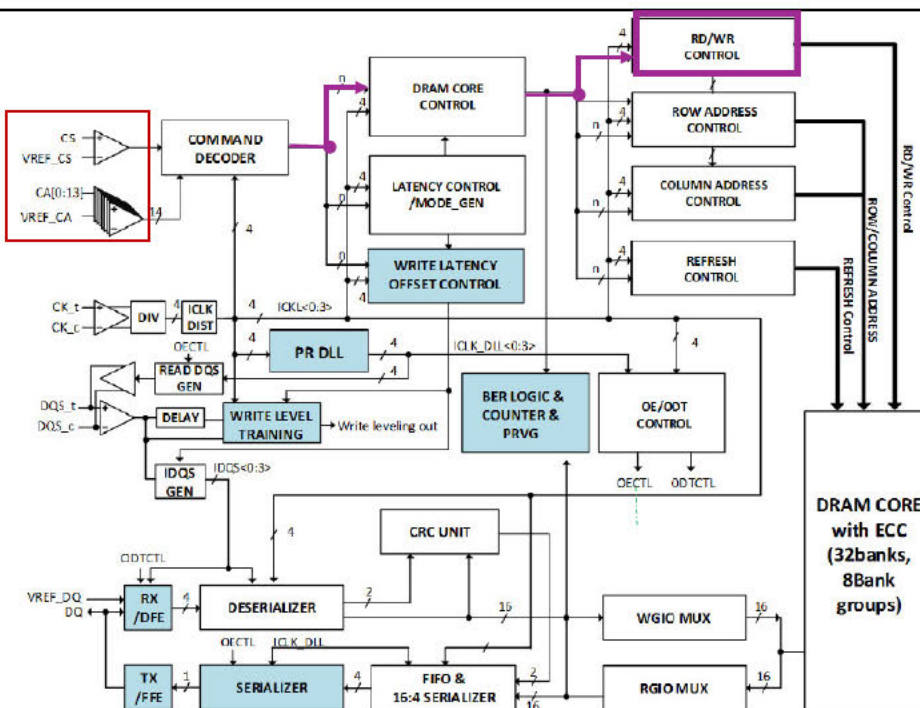


Fig. 2. DDR5 functional block diagram.

[1.g] [ii] and the data module in the second mode is configured to isolate the memory devices from being accessed by the system memory controller

In SK hynix's MRDIMMs, in the second (training) mode, the data module, comprised of the collection of the ten MDBs, is configured to isolate the memory devices from being accessed by the system memory controller.

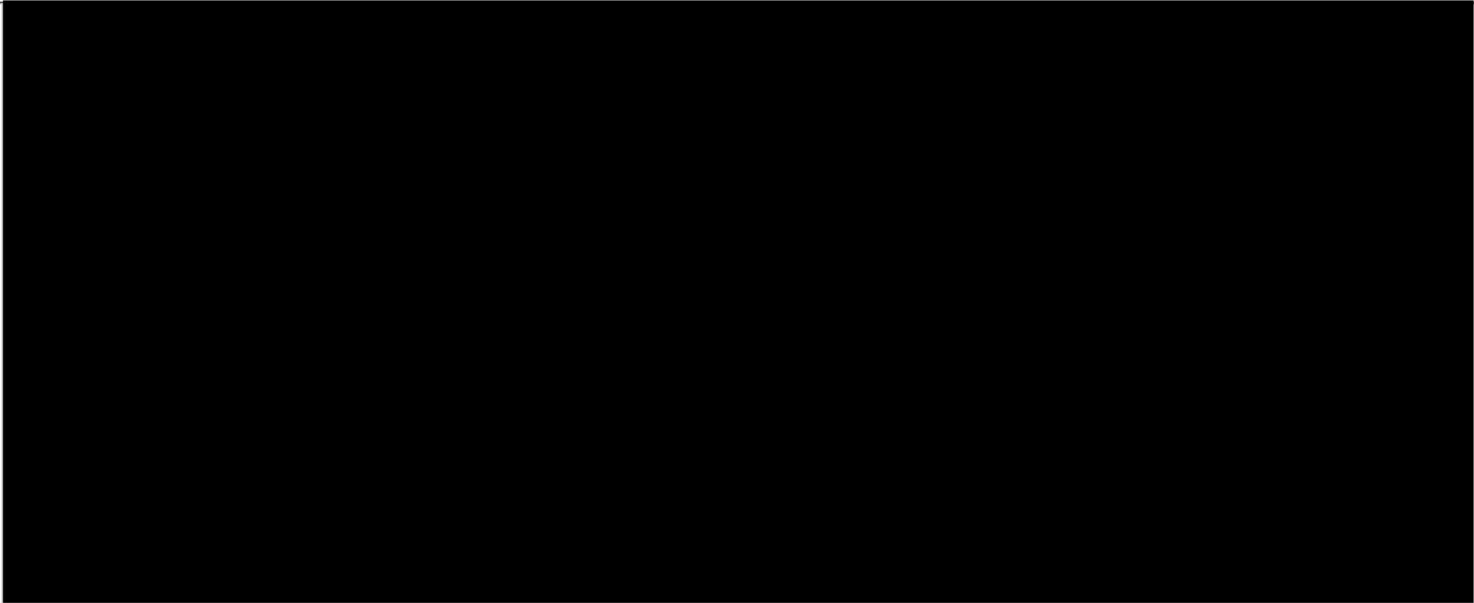
For example, in MRD training, while the host issues a command to read the DRAM's training pattern, what is being driven to the host is the result of the MDB's data comparison of the expected data pattern and the actual data pattern. See Ex. CC-68 (MDB01 Spec. Rev. 0.7) at 112 ("In this training mode, the data buffer uses a data pattern comparator to determine if the read data from the DRAMs matches an expected result. . . . The output of the data comparator is

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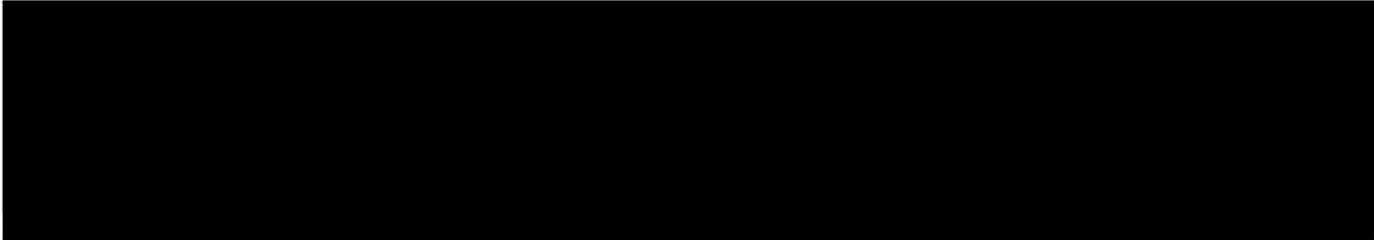
	driven onto the host connector interface on DQ pins.”). This indicates that the system memory controller is not accessing the content of the memory devices, but is instead only receiving results of MDB’s data pattern comparator. Similarly, in MWD, although the host issues write commands to the DRAMs, the data that is written is generated from the MDB’s pattern generator and not from the host; additionally, the read operation does not result in the data read from the DRAMs being sent to the host. Rather the data comparison results are sent to the host. <i>See id.</i> at 115 (“The host issues write commands to one selected rank in the Rank mode, or one or both PS in the Mux mode. The MDB generates the write pattern for the target rank or PS (one or both) based on the configuration of its Read Training Pattern registers and based on the PS and rank information from the write command. . . . The host then performs a read to the rank, or one or both PS and compares the received pattern against the ‘previously sent’ pattern. . . . The results for the write vs. read comparisons produced by the MDB for each PS is asynchronously sent back to the host through the DQ pins.”), <i>id.</i> (“To check whether the writes were successful, the host reads the data back from the same DRAM location and the data buffer performs a bit wise comparison with the read comparison pattern. The feedback on the DQ pins to the host will indicate a mis-compare in the same way as defined for the MRD training mode.”). This again indicates that the memory devices are not accessed by the memory controller for either read or write. Additionally, “On the Host interface, while in MWD training mode, the MDB continuously enables its DQ drivers, and disables its DQ receivers and DQ termination DQ_RTT_PARK. Also, the Data Buffer disables its DQS drivers and receivers. It applies DQS_RTT_PARK continuously on DQS if enabled. CRC modes must be disabled.” <i>Id.</i> This indicates that the MDB is not receiving any data from the system memory controller, as the receivers are disabled, again indicating that the memory controller and the memory devices are isolated by the MDB.
[1.g] [iii] and to transmit one or more second data signals including data patterns provided by the data handler logic elements to the data ports of the memory devices according to	In the training mode, in each of SK hynix’s MRDIMMs, the data module (collection of MDBs) is configured to transmit one or more second data signals, including data patterns provided by the data handler logic elements, to the data ports of the memory devices according to one or more commands output from the control module. As noted above, [REDACTED] [REDACTED]

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one or more
commands output
from the control
module,



Ex. CC-18C (MDB01 Spec. Rev. 0.7) at 115 (annotated).





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[REDACTED]

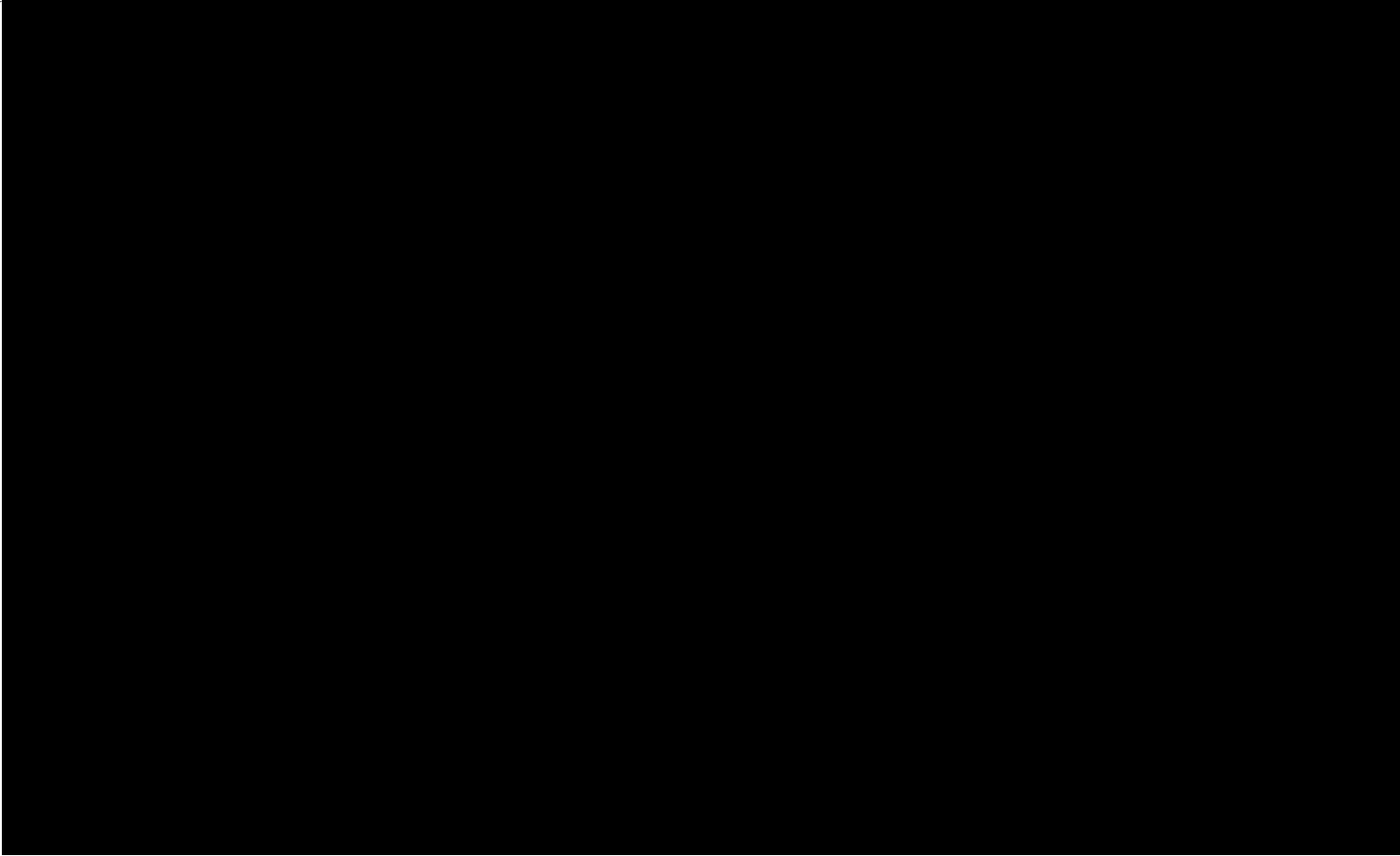
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	The host's write commands are first received by the MRCD and then output from the MRCD and sent to the MDB via the BCOM bus. <i>See</i> Ex. CC-22 (Montage M88MR5DB01) at 1 (The MDB has "[o]ne input-only control bus interface for connecting to a Multiplexed Rank Registering Clock Driver (MRCD)."); Ex. CC-12 (M88MR5RCD01 (Gen1 DDR5 MRCD) Flyer) at 1; Ex. CC-22 (M88MR5DB01 (Gen1 DDR5 MDB) Flyer) (MRCD has an "[i]ntegrated BCOM bus to control the multiplexed data buffers (MDB)."); <i>see also</i> Ex. CC-18C (MDB01 Spec. Rev. 0.7) at 114
[1.g] [iv] and wherein at least a portion of the memory devices are configured to receive the one or more second data signals according to the second memory	In the training mode, in each of SK hynix's MRDIMMs, at least a portion of the memory devices is configured to receive the one or more second data signals according to the second memory address and control signals from the control module. As noted above,



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address and control
signals from the
control module.



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Ex. CC-18C (MDB01 Spec. Rev. 0.7) at 115 (annotated).

Ex. CC-18C (MDB01 Spec. Rev. 0.7) at 115 (annotated).

See also Ex. CC-62 (Montage HK Proof) at 43 (“MRCD” is used to “buffer and re-drive clock, address, and command signals efficiently across multiple memory ranks . . .”):

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	“MRCD” multiplexed rank registering clock driver, a specialized buffer chip used in memory modules to buffer and re-drive clock, address, and command signals efficiently across multiple memory ranks, improving signal integrity and enabling higher memory speeds and capacities
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